AWS VPC PEERING

Student Project Report

Project Title: AWS VPC Peering – Private Connectivity Between Two VPCs

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Course: AWS Architect Associate

AWS Region: Asia Pacific (Hyderabad) – `ap-south-2`

Completion Date: 25/10/2026

1. Introduction

Amazon Virtual Private Cloud (Amazon VPC) enables users to create isolated virtual networks in AWS. VPC Peering establishes a private network connection between two VPCs, allowing resources in them to communicate using private IP addresses.

This project demonstrates how to configure two VPCs, create public and private subnets, configure route tables and security groups, establish VPC Peering, and verify private connectivity between EC2 instances.

2. Project Objectives

- Create two VPCs with non-overlapping IPv4 CIDR ranges.
- Configure public and private subnets.
- Deploy a bastion host and two private application instances.
- Establish a VPC Peering connection.
- Configure routes for forward and return traffic.
- Test connectivity using ICMP ping.
- Verify SSH access to App-B through the bastion host.

- Demonstrate controlled network failure and recovery.

3. AWS Services Used

AWS Service	Purpose
Amazon VPC	Provides isolated virtual networks
Amazon EC2	Hosts the bastion and application instances
VPC Peering	Connects VPC A and VPC B
Route Tables	Direct traffic to the correct destination
Internet Gateway	Provides internet connectivity for the bastion subnet
Security Groups	Control inbound and outbound instance traffic

4. Network Architecture

The project contains two VPCs connected through a VPC Peering connection.

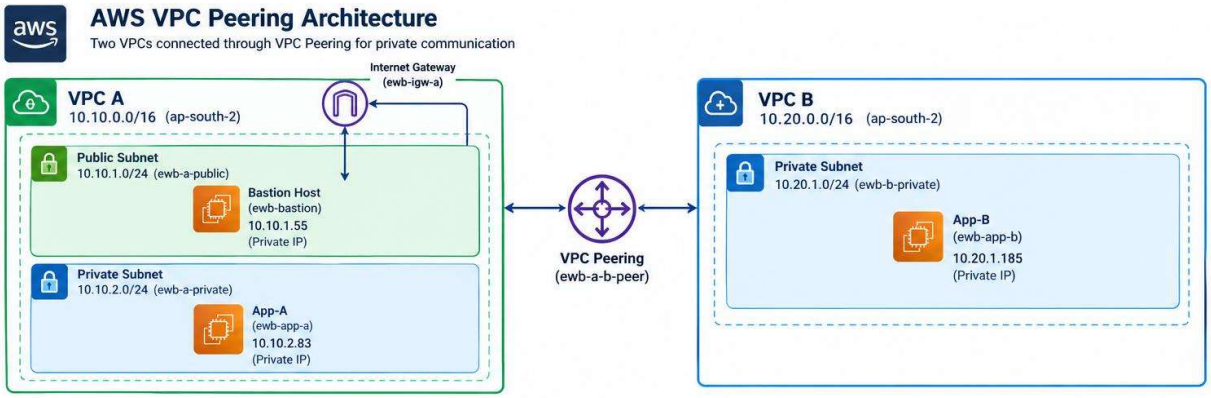
VPC A

- CIDR: `10.10.0.0/16`
- Public subnet: `10.10.1.0/24`
- Private subnet: `10.10.2.0/24`
- Bastion private IP: `10.10.1.55`
- App-A private IP: `10.10.2.83`

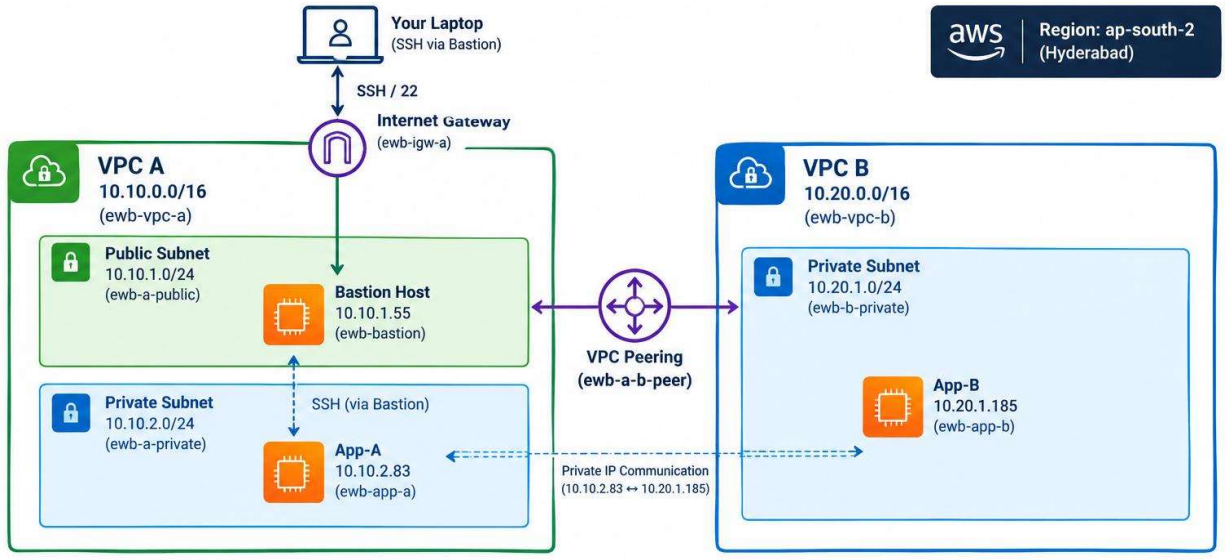
VPC B

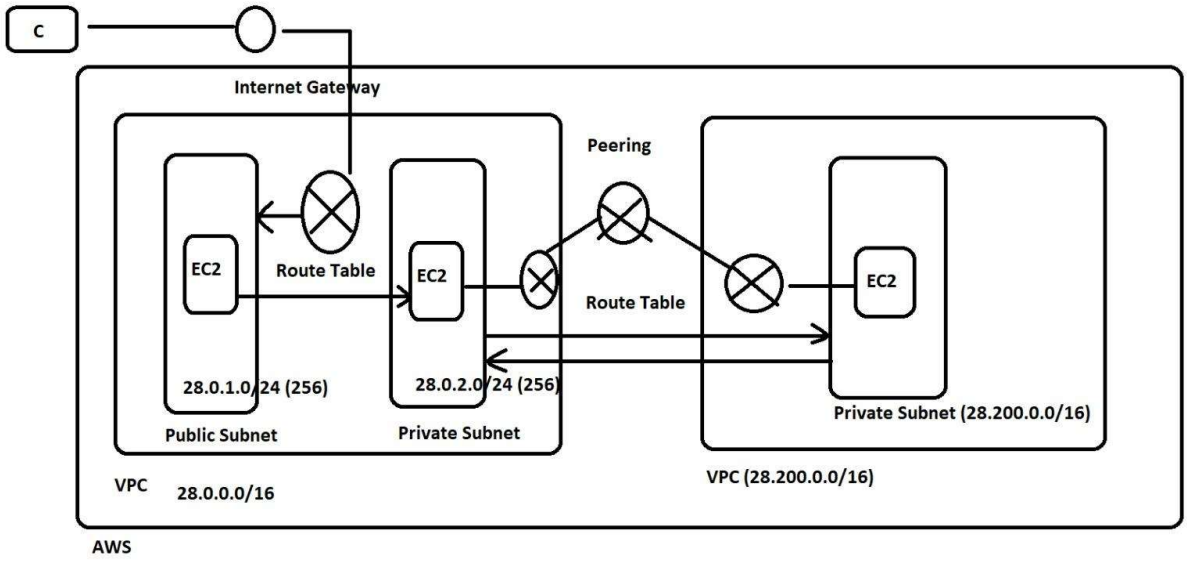
- CIDR: `10.20.0.0/16`
- Private subnet: `10.20.1.0/24`
- App-B private IP: `10.20.1.185`

The bastion host provides an administrative SSH path to the private instances. App-A and App-B communicate with each other using their private IPv4 addresses through VPC Peering.


AWS VPC Peering Architecture
Two VPCs connected through VPC Peering for private communication
Internet Gateway
(ewb-igw-a)
VPC A
10.10.0.0/16 (ap-south-2)
Public Subnet
10.10.1.0/24 (ewb-a-public)
Bastion Host
(ewb-bastion)
10.10.1.55
(Private IP)
Private Subnet
10.10.2.0/24 (ewb-a-private)
App-A
(ewb-app-a)
10.10.2.83
(Private IP)
VPC Peering
(ewb-a-b-peer)
VPC B
10.20.0.0/16 (ap-south-2)
Private Subnet
10.20.1.0/24 (ewb-b-private)
App-B
(ewb-app-b)
10.20.1.185
(Private IP)

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Your Laptop
(SSH via Bastion)
SSH / 22
Internet Gateway
(ewb-igw-a)
Region: ap-south-2
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VPC A
10.10.0.0/16
(ewb-vpc-a)
Public Subnet
10.10.1.0/24
(ewb-a-public)
Bastion Host
10.10.1.55
(ewb-bastion)
Private Subnet
10.10.2.0/24
(ewb-a-private)
SSH (via Bastion)
App-A
10.10.2.83
(ewb-app-a)
VPC Peering
(ewb-a-b-peer)
Private IP Communication
(10.10.2.83 ↔ 10.20.1.185)
VPC B
10.20.0.0/16
(ewb-vpc-b)
Private Subnet
10.20.1.0/24
(ewb-b-private)
App-B
10.20.1.185
(ewb-app-b)



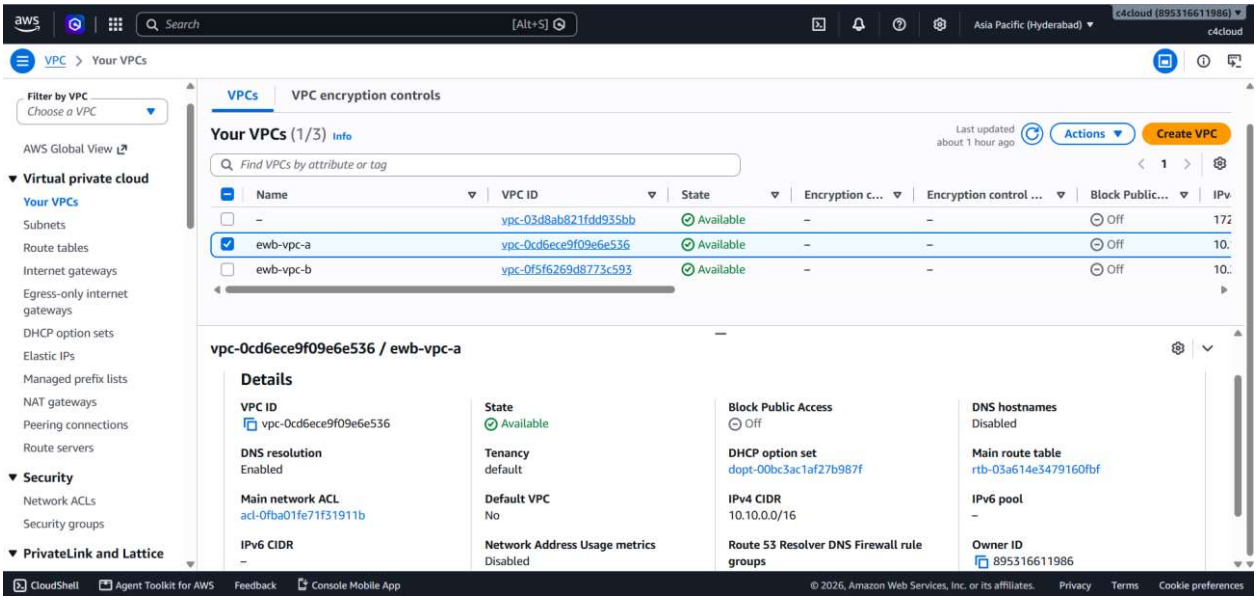
5. VPC and Subnet Configuration

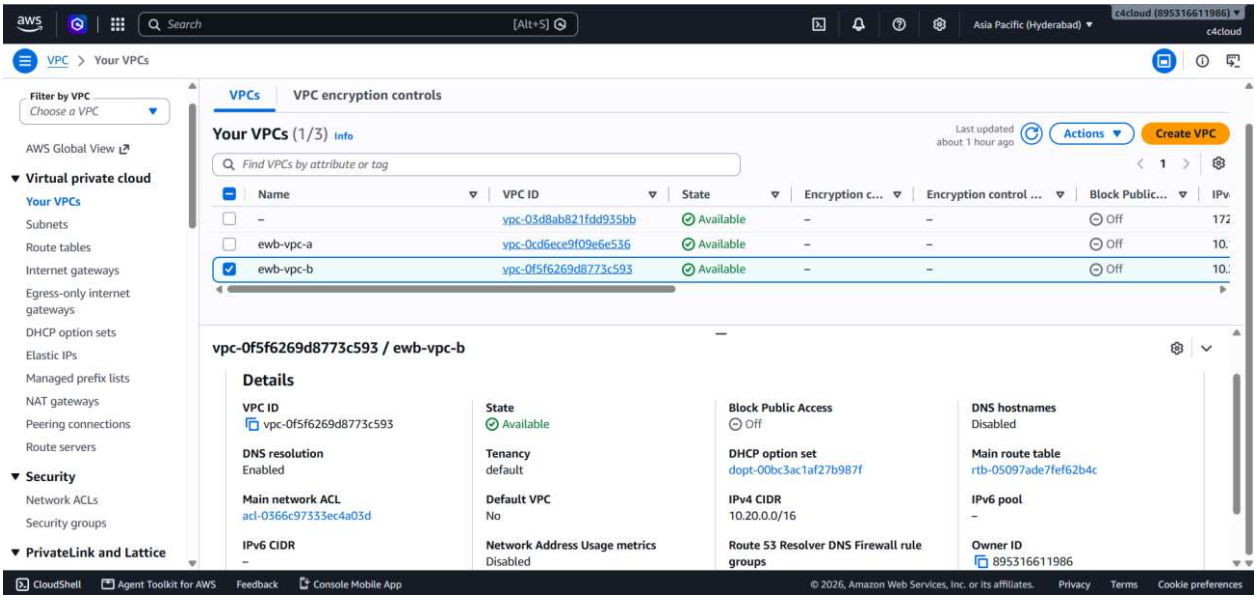
Two VPCs were configured with different CIDR ranges to avoid address overlap.

Resource	Name	CIDR
VPC A	`ewb-vpc-a`	`10.10.0.0/16`
Public subnet A	`ewb-a-public`	`10.10.1.0/24`
Private subnet A	`ewb-a-private`	`10.10.2.0/24`
VPC B	`ewb-vpc-b`	`10.20.0.0/16`
Private subnet B	`ewb-b-private`	`10.20.1.0/24`

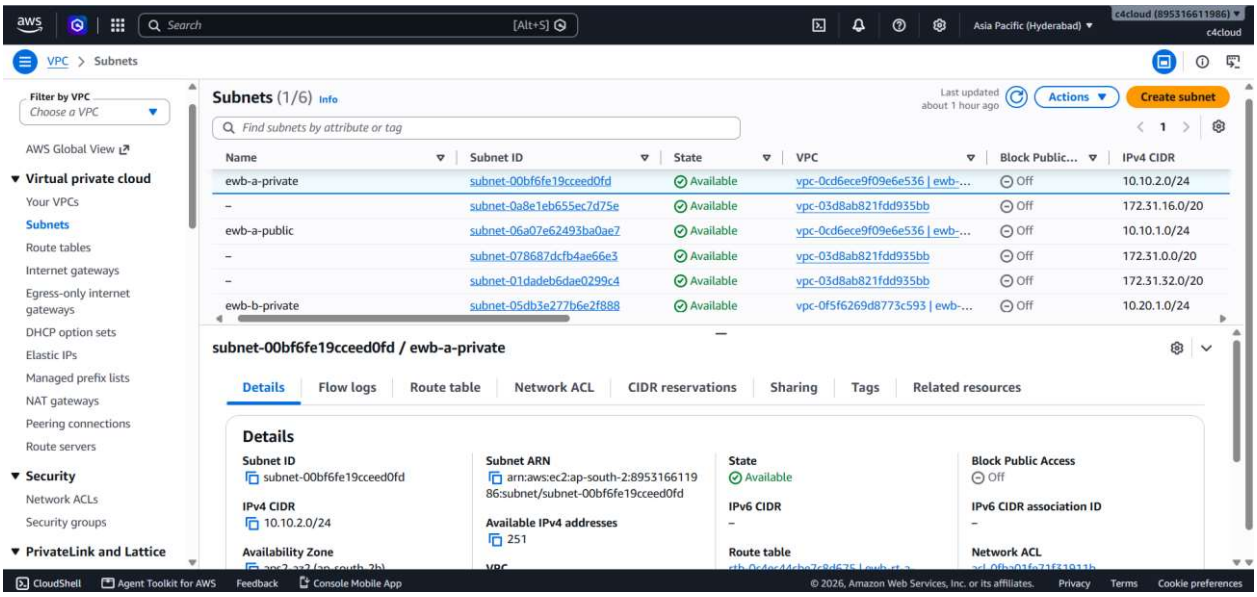
An Internet Gateway named `ewb-igw-a` was attached to VPC A. VPC B was configured without an Internet Gateway for this lab.

VPCs:

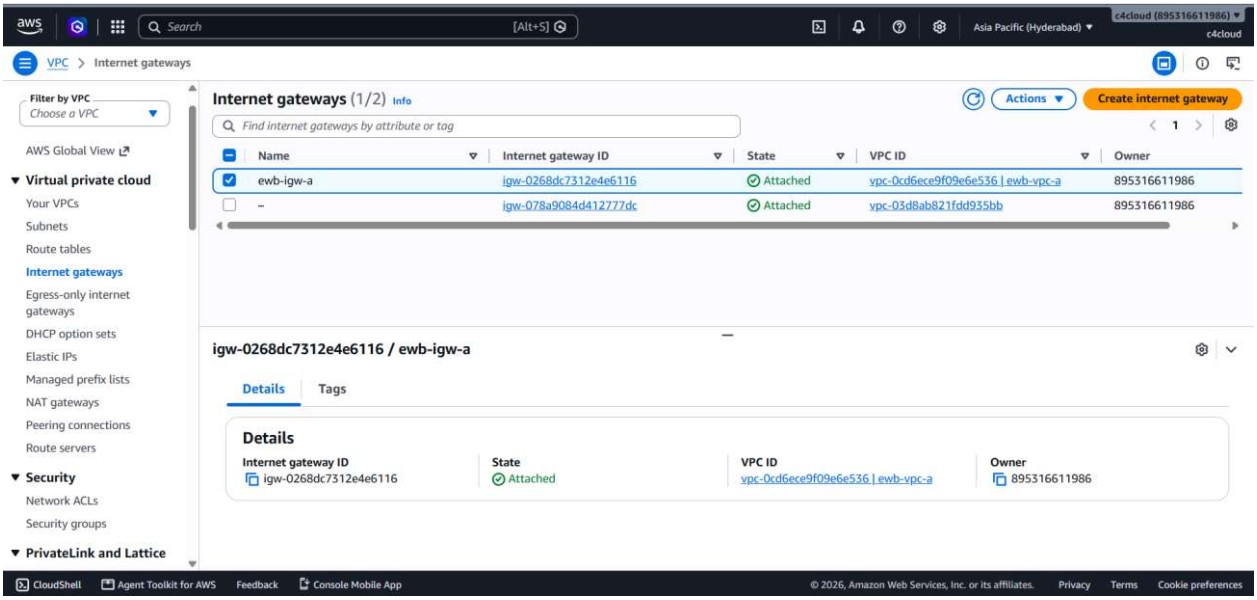

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Security groups
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Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv
– vpc-03d8ab821fdd935bb Available – – Off 172
ewb-vpc-a vpc-0cd6ece9f09e6e536 Available – – Off 10.
ewb-vpc-b vpc-0f5f6269d8773c593 Available – – Off 10.
vpc-0cd6ece9f09e6e536 / ewb-vpc-a
Details
VPC ID
vpc-0cd6ece9f09e6e536
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-00bc3ac1af27b987f
Main route table
rtb-03a614e3479160fbf
Main network ACL
acl-0fba01fe71f31911b
Default VPC
No
IPv4 CIDR
10.10.0.0/16
IPv6 pool
–
IPv6 CIDR
–
Network Address Usage metrics
Disabled
Route 53 Resolver DNS Firewall rule groups
Owner ID
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Your VPCs (1/3) Info
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Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv
– vpc-03d8ab821fdd935bb Available – – Off 172
ewb-vpc-a vpc-0cd6ece9f09e6e536 Available – – Off 10.
ewb-vpc-b vpc-0f5f6269d8773c593 Available – – Off 10.
vpc-0f5f6269d8773c593 / ewb-vpc-b
Details
VPC ID
vpc-0f5f6269d8773c593
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-00bc3ac1af27b987f
Main route table
rtb-05097ade7fef62b4c
Main network ACL
acl-0366c97333ec4a03d
Default VPC
No
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10.20.0.0/16
IPv6 pool
–
IPv6 CIDR
–
Network Address Usage metrics
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Subnet:



Internet Gateway:



6. EC2 Instance Configuration

Three Amazon Linux 2023 EC2 instances were used.

Instance	Private IP	Role
`ewb-bastion`	`10.10.1.55`	Administrative access
`ewb-app-a`	`10.10.2.83`	Application host in VPC A
`ewb-app-b`	`10.20.1.185`	Application host in VPC B

The bastion host was placed in the public subnet. App-A and App-B were placed in their respective private subnets and do not require public IPv4 addresses.

Bastion Host Configuration

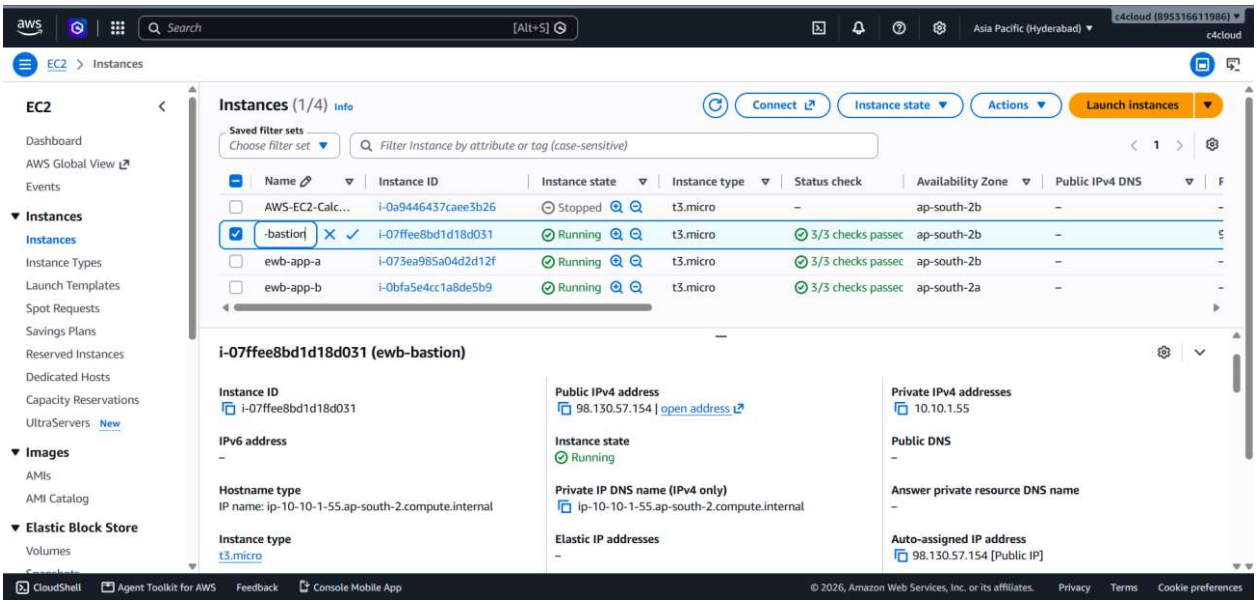
The `ewb-bastion` instance was deployed in the public subnet of VPC A, with private IP address `10.10.1.55`. It was used as an administrative entry point for accessing the application instances in the private subnets.

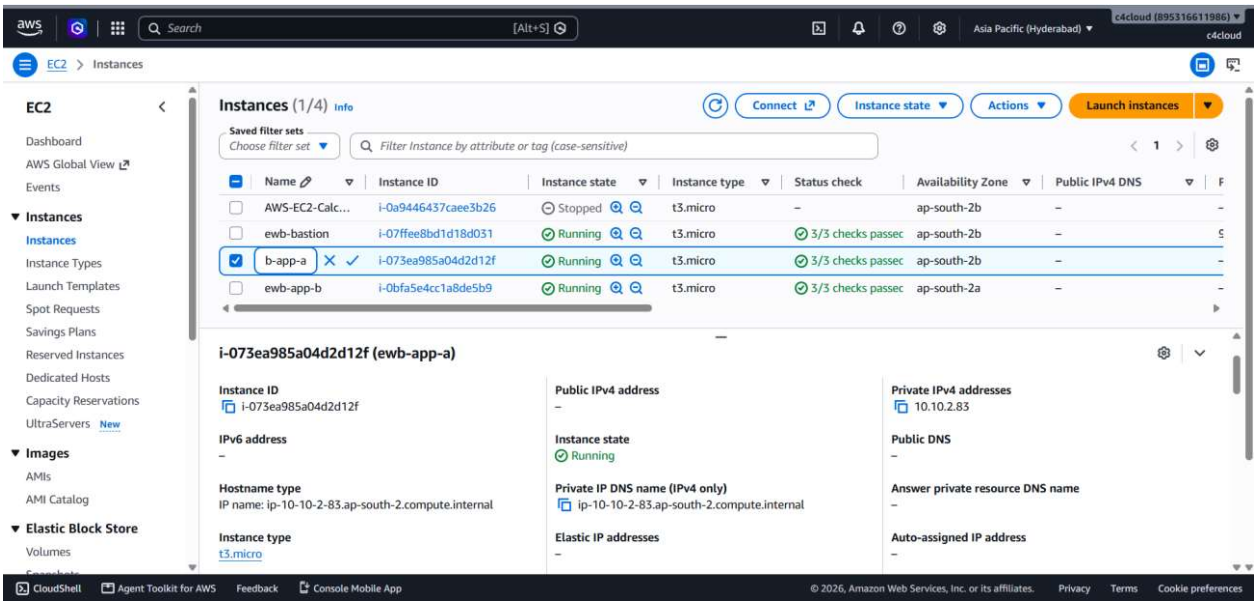
SSH access to the bastion host was configured using a security group rule that allowed access from the student's public IP address using a `/32` CIDR. This restricted inbound SSH access to the specified source address rather than allowing SSH connections from the entire internet.

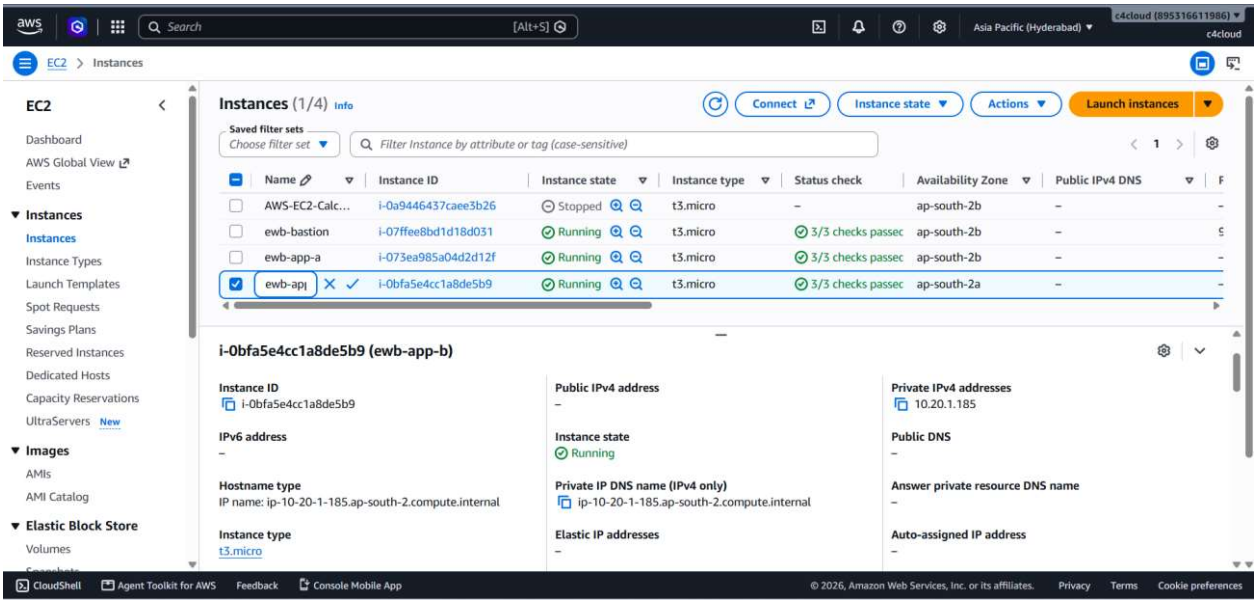
The bastion host provided a way to reach private instances without requiring public IPv4 addresses on those instances.

Private Application Instances

The `ewb-app-a` instance was deployed in the private subnet of VPC A with private IP address `10.10.2.83`. The `ewb-app-b` instance was deployed in the private subnet of VPC B with private IP address `10.20.1.185`.


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Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
AWS-EC2-Calc...
i-0a9446437caee3b26
Stopped
t3.micro
–
ap-south-2b
–
-bastion
i-07ffee8bd1d18d031
Running
t3.micro
3/3 checks passed
ap-south-2b
–
ewb-app-a
i-073ea985a04d2d12f
Running
t3.micro
3/3 checks passed
ap-south-2b
–
ewb-app-b
i-0bfa5e4cc1a8de5b9
Running
t3.micro
3/3 checks passed
ap-south-2a
–
i-07ffee8bd1d18d031 (ewb-bastion)
Instance ID
i-07ffee8bd1d18d031
IPv6 address
–
Hostname type
IP name: ip-10-10-1-55.ap-south-2.compute.internal
Instance type
t3.micro
Public IPv4 address
98.130.57.154 | open address
Instance state
Running
Private IP DNS name (IPv4 only)
ip-10-10-1-55.ap-south-2.compute.internal
Elastic IP addresses
–
Private IPv4 addresses
10.10.1.55
Public DNS
–
Answer private resource DNS name
–
Auto-assigned IP address
98.130.57.154 [Public IP]
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7. Route Table Configuration

The route tables were configured to provide connectivity within each VPC and between the peered VPCs.

VPC A – Private Route Table (`ewb-rt-a-private`)

- `10.10.0.0/16` → local
- `10.20.0.0/16` → VPC Peering connection

VPC B – Private Route Table (`ewb-rt-b-private`)

- `10.20.0.0/16` → local
- `10.10.0.0/16` → VPC Peering connection

VPC A – Public Route Table (`ewb-rt-a-public`)

- `10.10.0.0/16` → local
- `0.0.0.0/0` → Internet Gateway
- `10.20.0.0/16` → VPC Peering connection

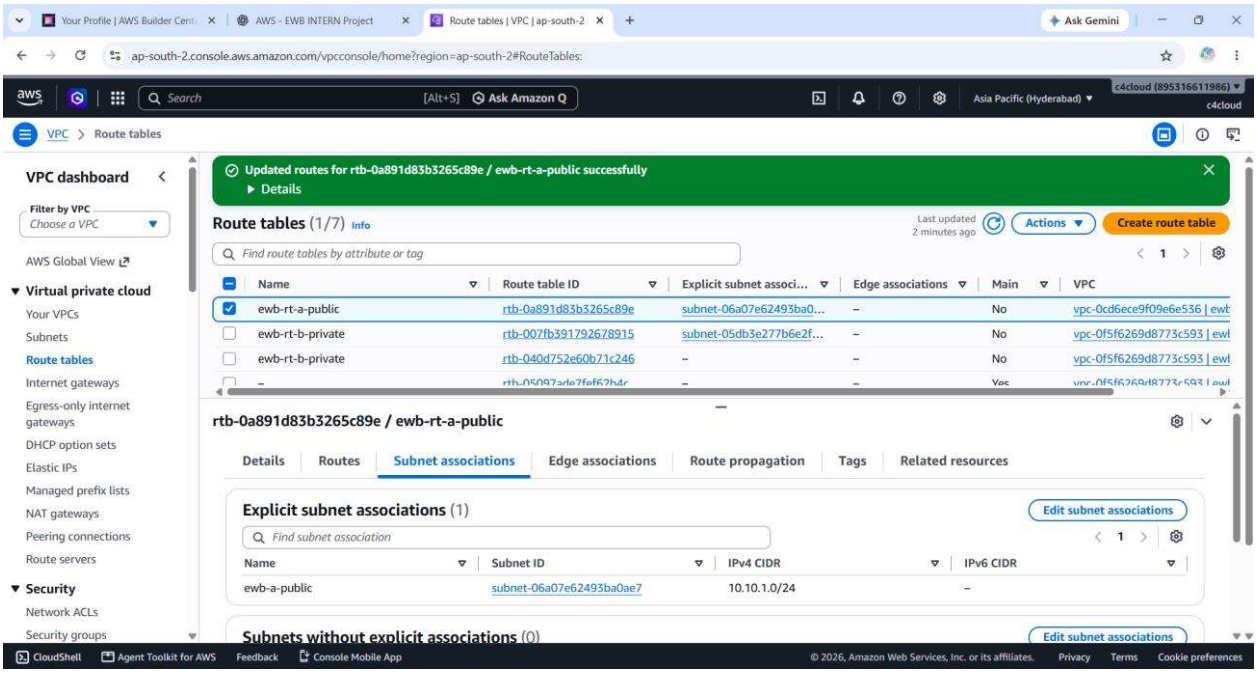
The peering routes provide the forward and return paths needed for private communication between the two application instances.

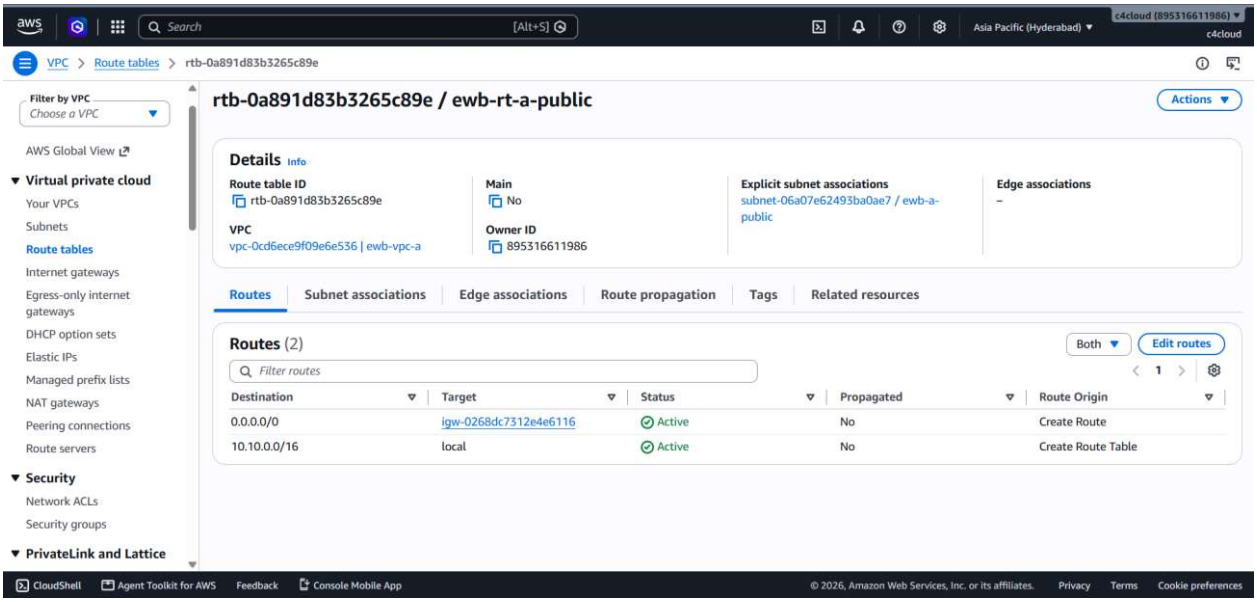
1. VPC A — Public Route Table (`ewb-rt-a-public`)

This route table is associated with the public subnet in `ewb-vpc-a`, where the bastion host is deployed.

Destination	Target	Purpose
`10.10.0.0/16`	Local	Communication within VPC A
`0.0.0.0/0`	Internet Gateway (`ewb-igw-a`)	Outbound internet routing
`10.20.0.0/16`	VPC Peering (`ewb-a-b-peer`)	Traffic from VPC A toward VPC B

The local route enables communication within VPC A. The Internet Gateway route provides an internet path for eligible resources in the public subnet, subject to their public IP configuration and security rules. The peering route directs traffic destined for VPC B through the peering connection.





Route Table Configuration Details

Three route tables were used for the project.

1. VPC A — Public Route Table (`ewb-rt-a-public`)

This route table is associated with the public subnet in `ewb-vpc-a`, where the bastion host is deployed.

Destination	Target	Purpose

`10.10.0.0/16`	Local	Communication within VPC A
`0.0.0.0/0`	Internet Gateway (`ewb-igw-a`)	Outbound internet routing
`10.20.0.0/16`	VPC Peering (`ewb-a-b-peer`)	Traffic from VPC A toward VPC B

The local route enables communication within VPC A. The Internet Gateway route provides an internet path for eligible resources in the public subnet, subject to their public IP configuration and security rules. The peering route directs traffic destined for VPC B through the peering connection.

2. VPC A — Private Route Table (`ewb-rt-a-private`)

This route table is associated with the private subnet containing App-A.

Destination	Target	Purpose
`10.10.0.0/16`	Local	Communication within VPC A
`10.20.0.0/16`	VPC Peering (`ewb-a-b-peer`)	Communication from App-A toward VPC B

The peering route allows App-A, with private IP address `10.10.2.83`, to send traffic to App-B in VPC B.

3. VPC B — Private Route Table (`ewb-rt-b-private`)

This route table is associated with the private subnet containing App-B.

Destination	Target	Purpose
`10.20.0.0/16`	Local	Communication within VPC B
`10.10.0.0/16`	VPC Peering (`ewb-a-b-peer`)	Return and other traffic toward VPC A

This route allows App-B, with private IP address `10.20.1.185`, to send traffic to App-A in VPC A.

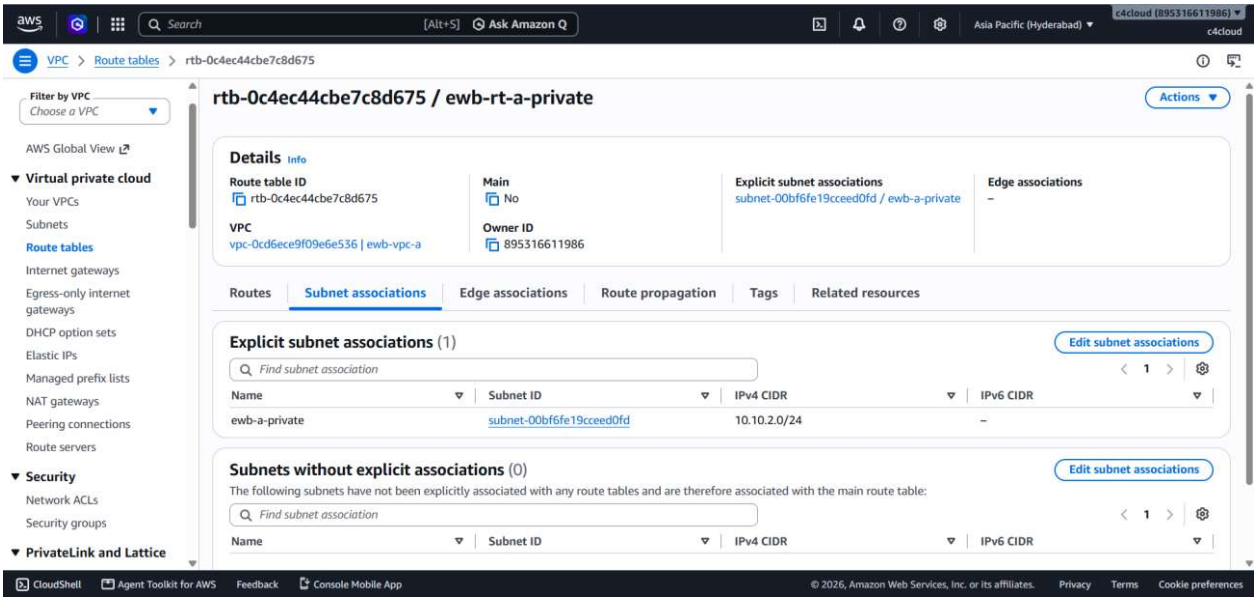
Connectivity Testing

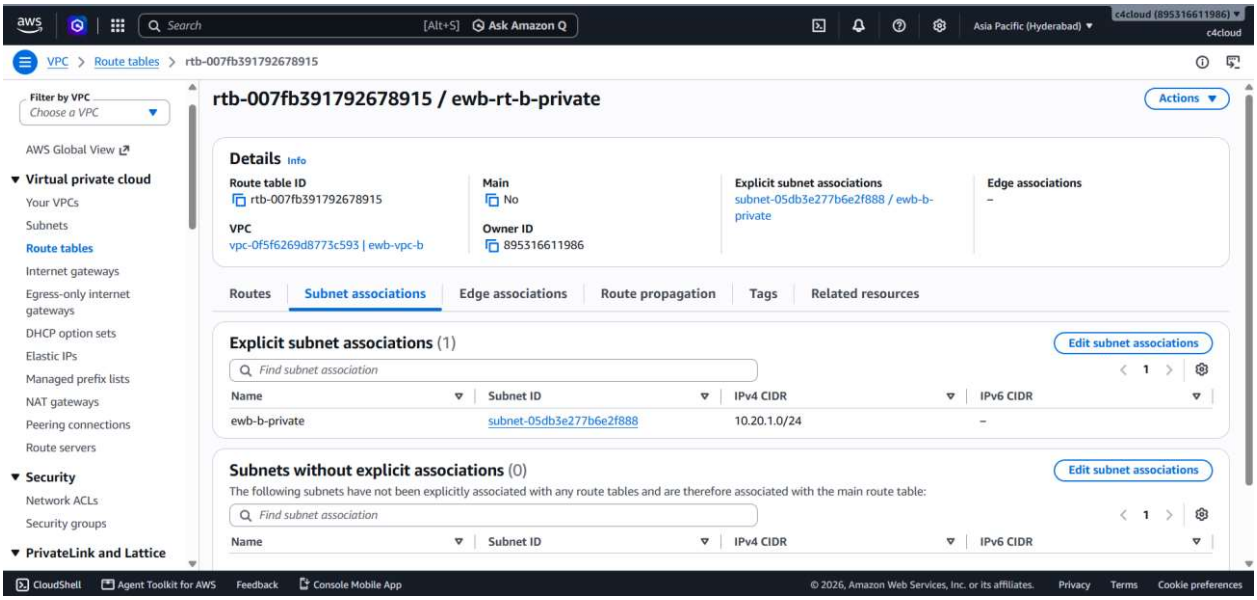
After configuring the routes, connectivity was verified using ICMP ping tests.

- **Forward test:** App-A (`10.10.2.83`) to App-B (`10.20.1.185`) — four packets received with 0% packet loss.

- **Reverse test:** App-B (`10.20.1.185`) to App-A (`10.10.2.83`) — four packets received with 0% packet loss.

These results confirmed that the configured routes and applicable network controls supported communication between the two application instances during testing.

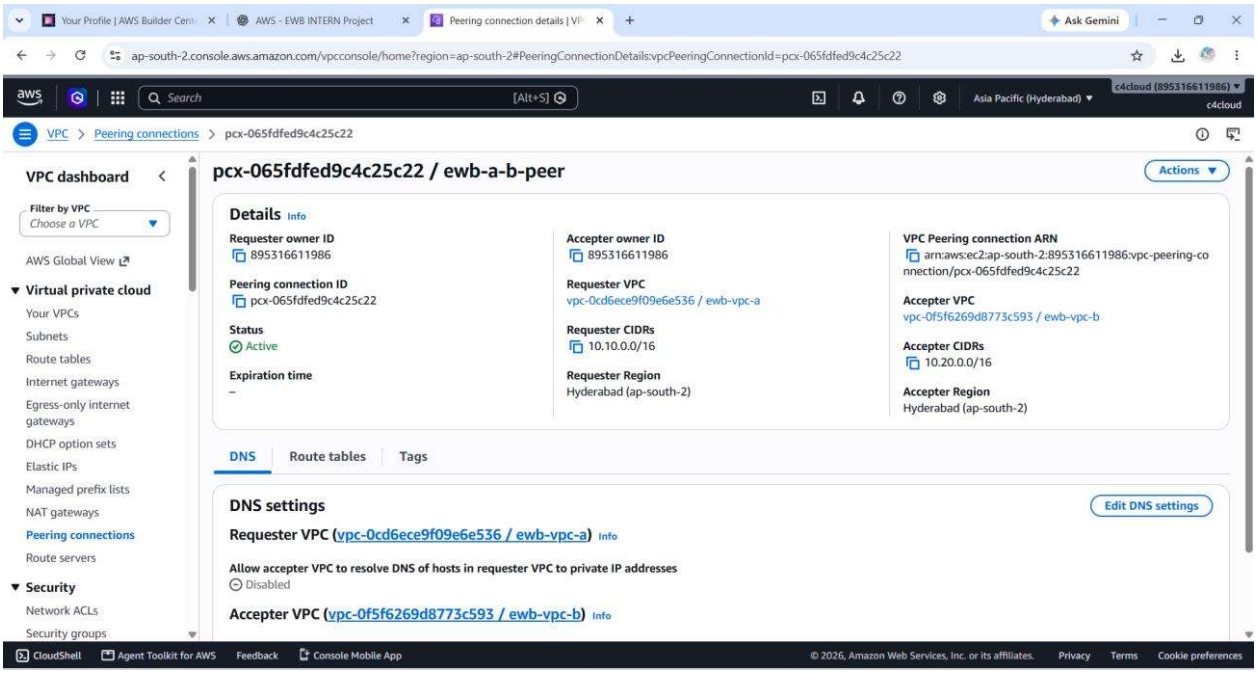


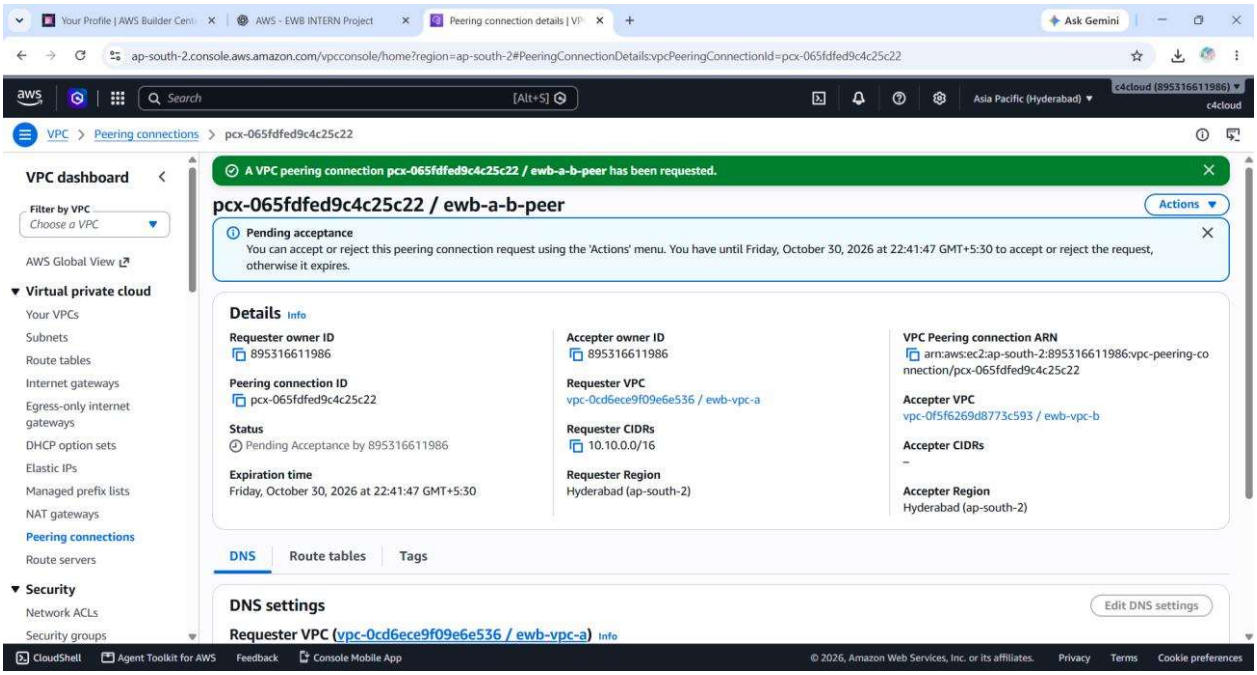


8. VPC Peering Configuration

A VPC Peering connection named `ewb-a-b-peer` was created between `ewb-vpc-a` and `ewb-vpc-b`.

The connection was accepted and reached the Active state. Routes were then configured in the associated route tables to direct traffic to the remote VPC through the peering connection.



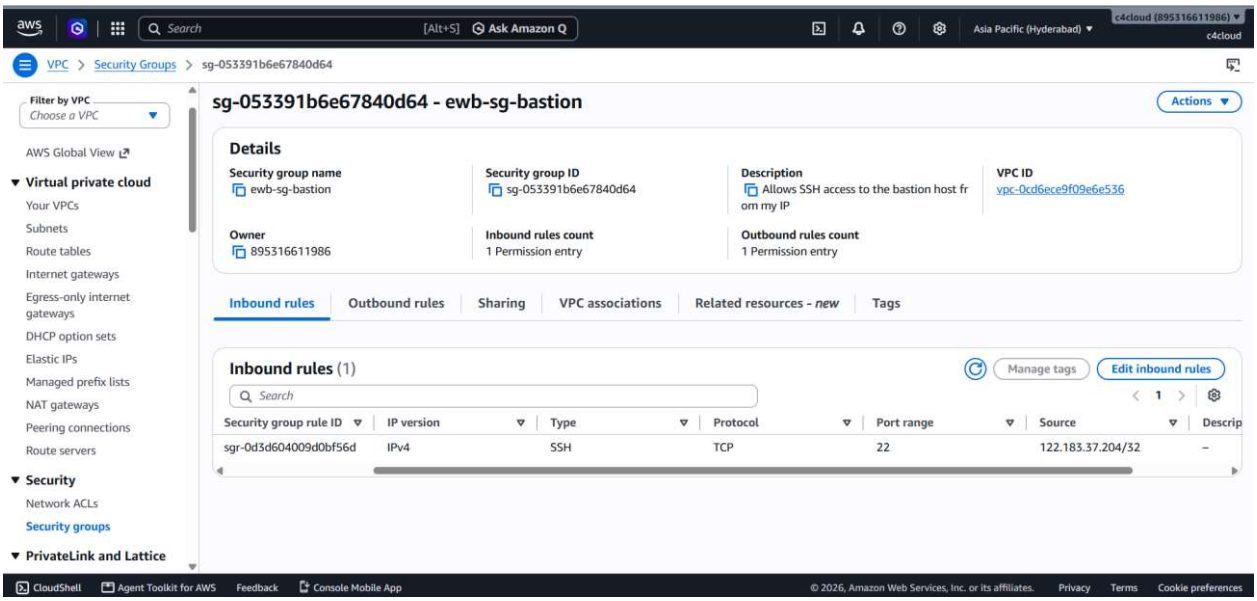


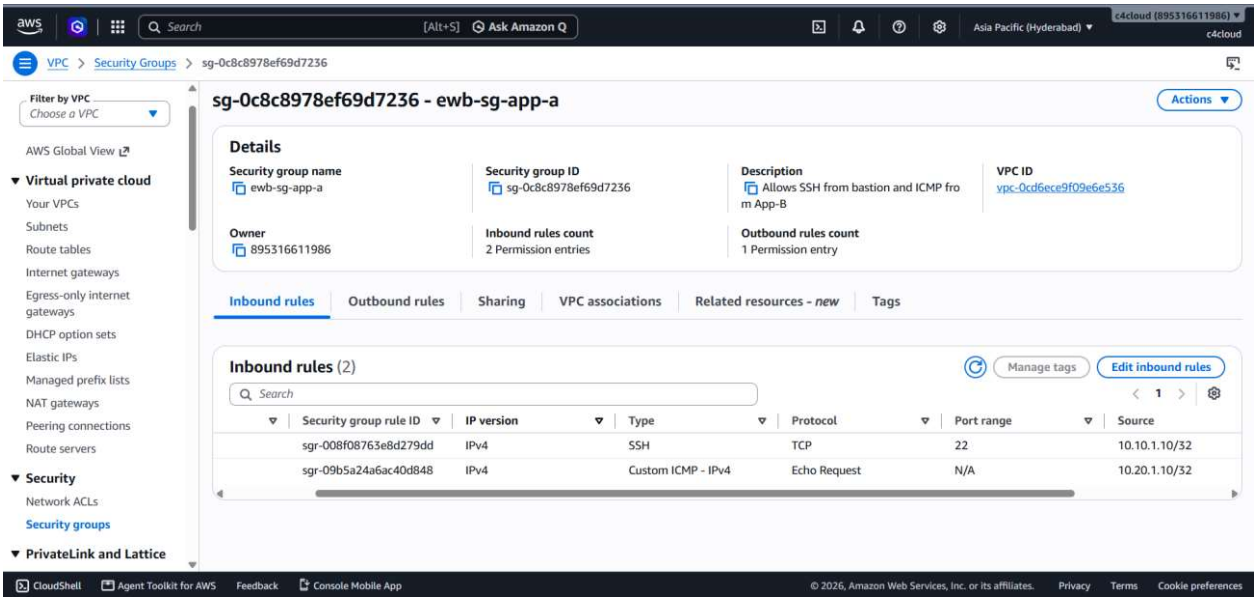
9. Security Group Configuration

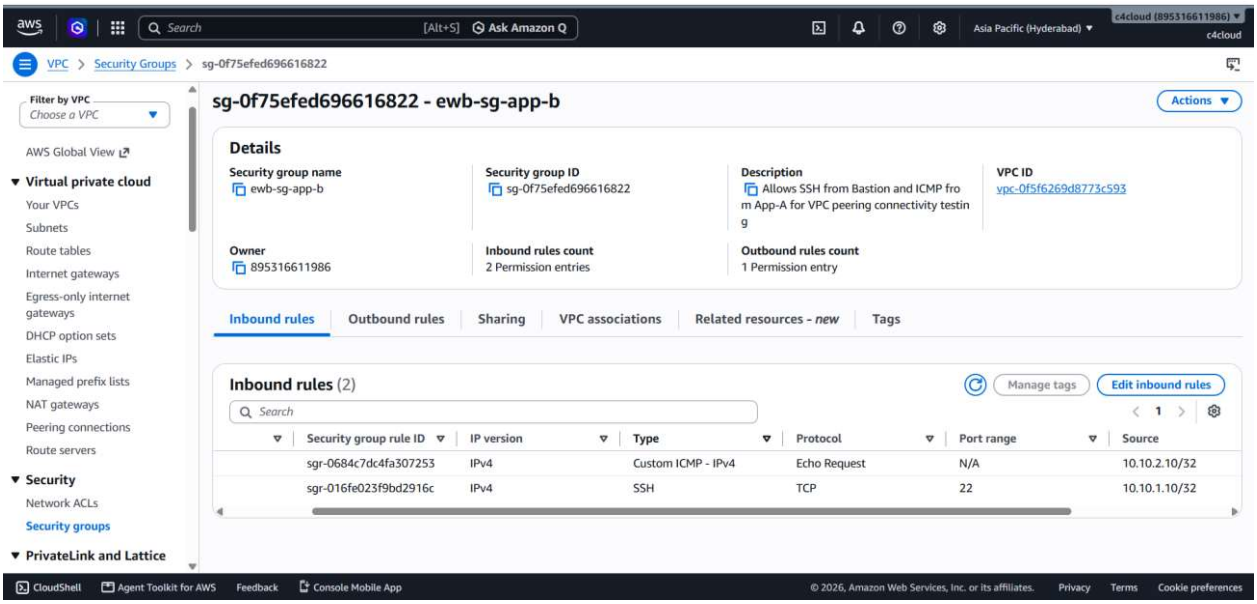
Security groups were configured to allow only the required administrative and connectivity-testing traffic.

- Bastion: SSH access from the student's current public IP address using a /32 rule.
- App-A: SSH access from the bastion private IP and ICMP testing from App-B.
- App-B: SSH access from the bastion private IP and ICMP testing from App-A.

The ICMP rules were configured using the actual private IP addresses of the application instances. SSH access was not opened to all internet addresses.


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Security
Network ACLs
Security groups
PrivateLink and Lattice
sg-053391b6e67840d64 - ewb-sg-bastion
Actions
Details
Security group name
ewb-sg-bastion
Security group ID
sg-053391b6e67840d64
Description
Allows SSH access to the bastion host from my IP
VPC ID
vpc-0cd6ece9f09e6e536
Owner
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Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Security group rule ID
IP version
Type
Protocol
Port range
Source
Descrip
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IPv4
SSH
TCP
22
122.183.37.204/32
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Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
sg-0c8c8978ef69d7236 - ewb-sg-app-a
Actions
Details
Security group name
ewb-sg-app-a
Security group ID
sg-0c8c8978ef69d7236
Description
Allows SSH from bastion and ICMP from App-B
VPC ID
vpc-0cd6ece9f09e6e536
Owner
895316611986
Inbound rules count
2 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Security group rule ID
IP version
Type
Protocol
Port range
Source
sgr-008f08763e8d279dd
IPv4
SSH
TCP
22
10.10.1.10/32
sgr-09b5a24a6ac40d848
IPv4
Custom ICMP - IPv4
Echo Request
N/A
10.20.1.10/32
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



10. Connectivity Testing

10.1 Forward Connectivity: App-A to App-B

Source: App-A (`10.10.2.83`)

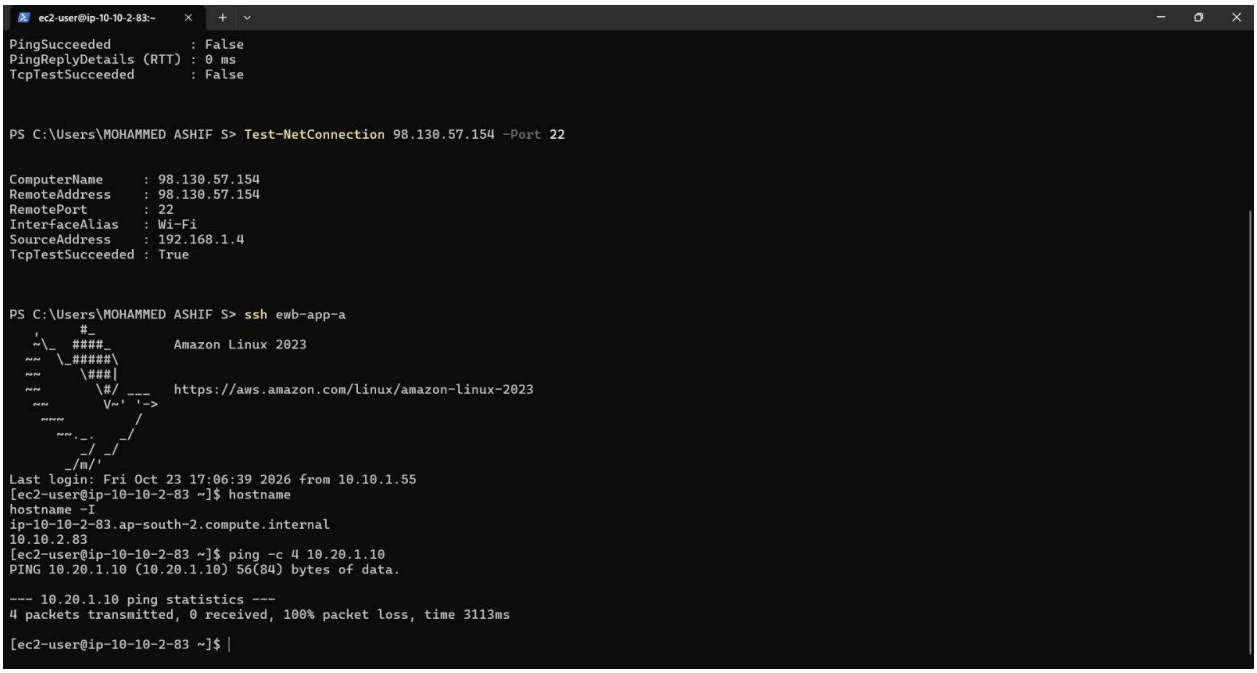
Destination: App-B (`10.20.1.185`)

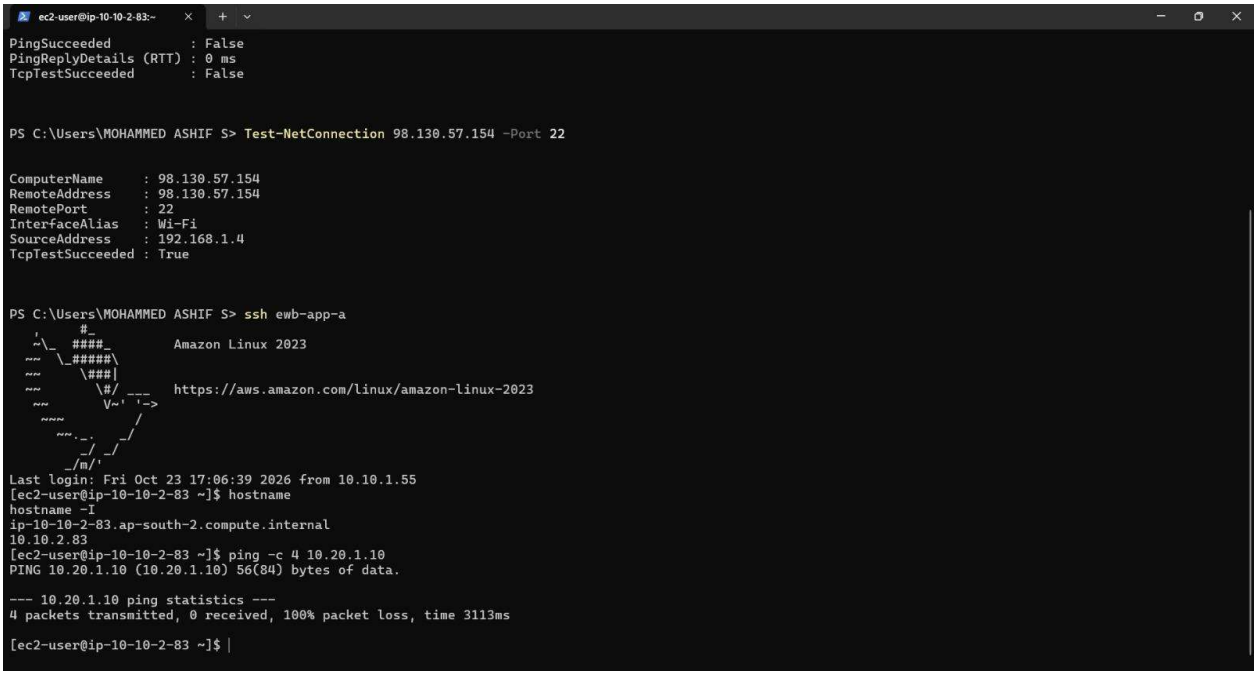
Command:

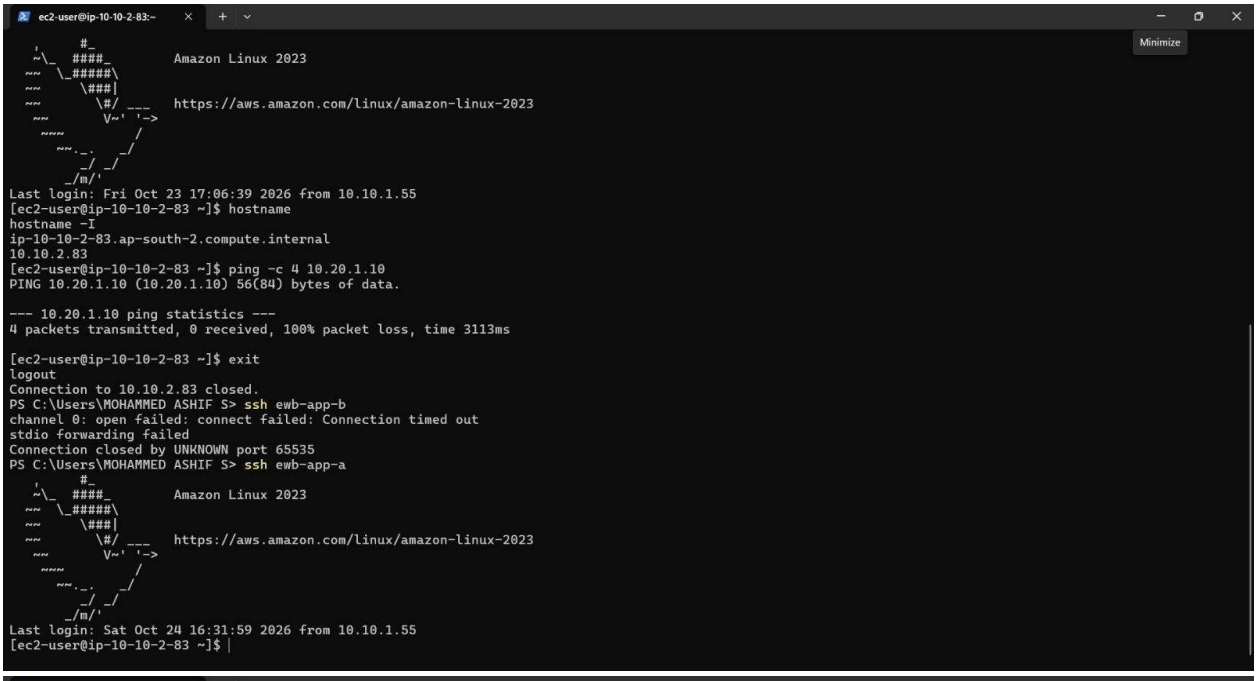
```
ping -c 4 10.20.1.185
```

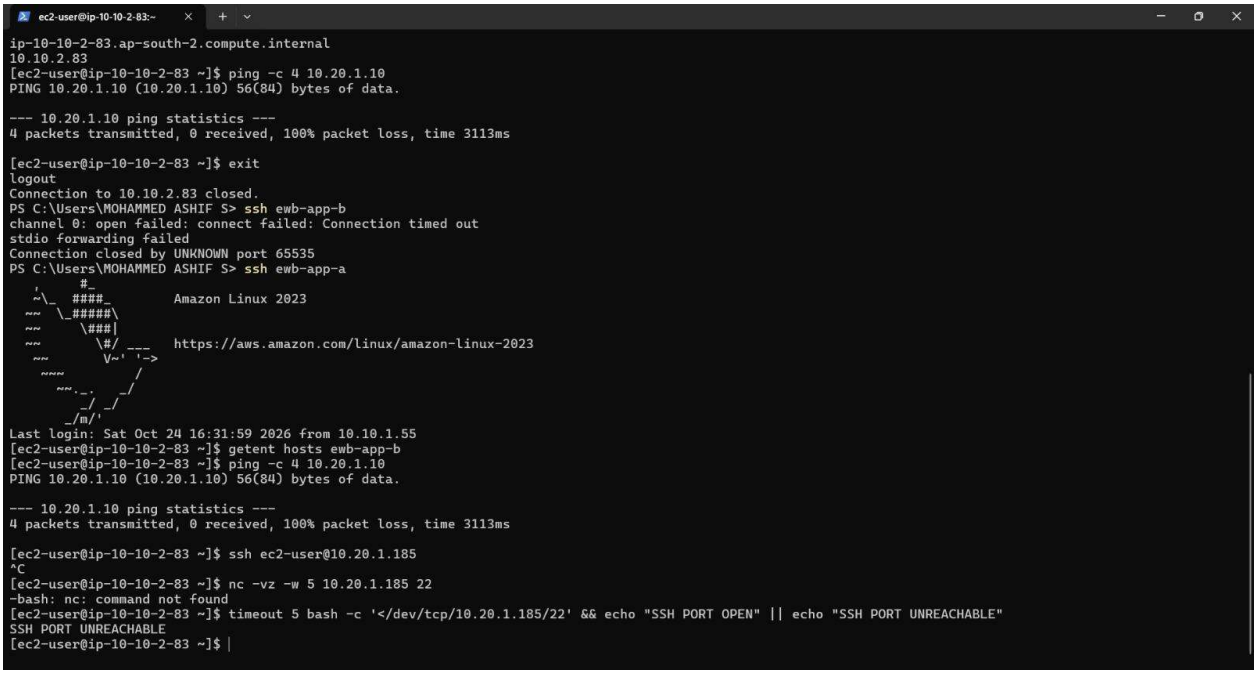
Observed result: Four packets were transmitted, four packets were received, and packet loss was 0%.

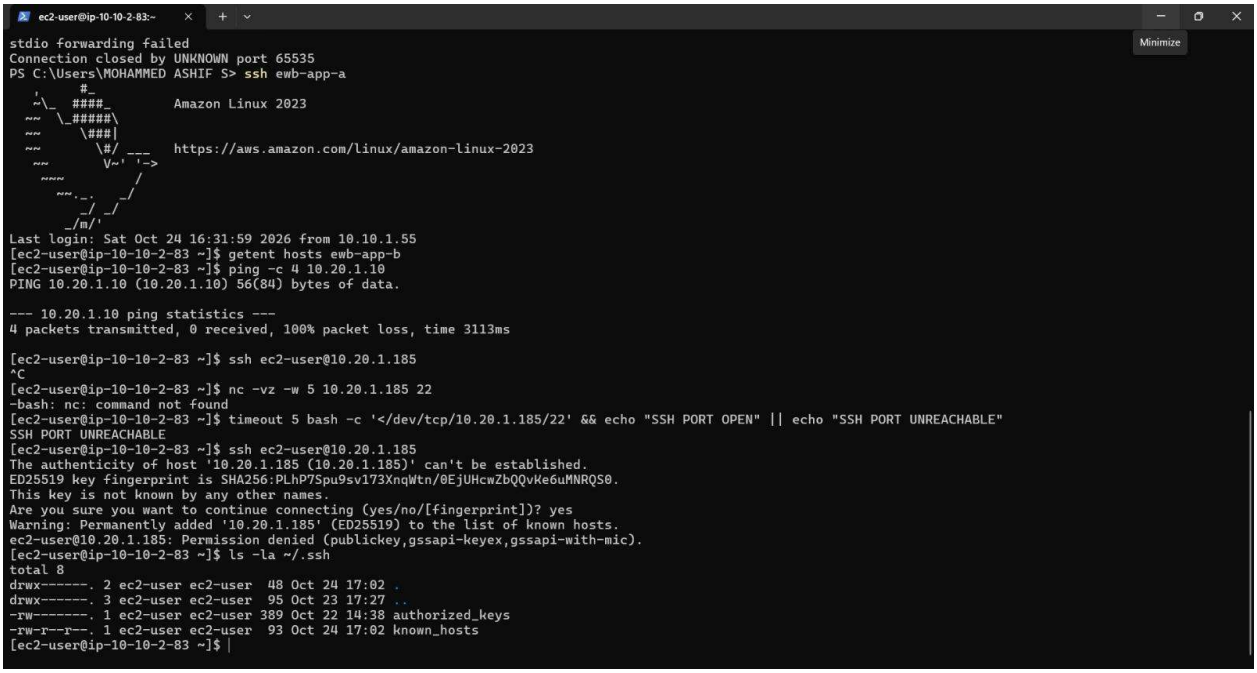
Conclusion: Forward private connectivity was successful.


ec2-user@ip-10-10-2-83:~
PingSucceeded : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded : False
PS C:\Users\MOHAMMED ASHIF S> Test-NetConnection 98.130.57.154 -Port 22
ComputerName : 98.130.57.154
RemoteAddress : 98.130.57.154
RemotePort : 22
InterfaceAlias : Wi-Fi
SourceAddress : 192.168.1.4
TcpTestSucceeded : True
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-a
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Fri Oct 23 17:06:39 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ hostname
hostname -I
ip-10-10-2-83.ap-south-2.compute.internal
10.10.2.83
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3113ms
[ec2-user@ip-10-10-2-83 ~]$


ec2-user@ip-10-10-2-83:~
PingSucceeded : False
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded : False
PS C:\Users\MOHAMMED ASHIF S> Test-NetConnection 98.130.57.154 -Port 22
ComputerName : 98.130.57.154
RemoteAddress : 98.130.57.154
RemotePort : 22
InterfaceAlias : Wi-Fi
SourceAddress : 192.168.1.4
TcpTestSucceeded : True
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-a
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Fri Oct 23 17:06:39 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ hostname
hostname -I
ip-10-10-2-83.ap-south-2.compute.internal
10.10.2.83
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3113ms
[ec2-user@ip-10-10-2-83 ~]$


ec2-user@ip-10-10-2-83:~
Minimize
#_
~_ ####_ Amazon Linux 2023
~~ _#####\
~~ \###|
~~ \#/ ___ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~ /
~~._. _/
_/ _/
_/m/'
Last login: Fri Oct 23 17:06:39 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ hostname
hostname -I
ip-10-10-2-83.ap-south-2.compute.internal
10.10.2.83
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3113ms
[ec2-user@ip-10-10-2-83 ~]$ exit
logout
Connection to 10.10.2.83 closed.
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-b
channel 0: open failed: connect failed: Connection timed out
stdio forwarding failed
Connection closed by UNKNOWN port 65535
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-a
#_
~_ ####_ Amazon Linux 2023
~~ _#####\
~~ \###|
~~ \#/ ___ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~ /
~~._. _/
_/ _/
_/m/'
Last login: Sat Oct 24 16:31:59 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$


ec2-user@ip-10-10-2-83:~
ip-10-10-2-83.ap-south-2.compute.internal
10.10.2.83
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3113ms
[ec2-user@ip-10-10-2-83 ~]$ exit
logout
Connection to 10.10.2.83 closed.
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-b
channel 0: open failed: connect failed: Connection timed out
stdio forwarding failed
Connection closed by UNKNOWN port 65535
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-a
#_
~_ ####_ Amazon Linux 2023
~~ _#####\
~~ \###|
~~ \#/ ___ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~ /
~~._. _/
_/ _/
_/m/'
Last login: Sat Oct 24 16:31:59 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ getent hosts ewb-app-b
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3113ms
[ec2-user@ip-10-10-2-83 ~]$ ssh ec2-user@10.20.1.185
^C
[ec2-user@ip-10-10-2-83 ~]$ nc -vz -w 5 10.20.1.185 22
-bash: nc: command not found
[ec2-user@ip-10-10-2-83 ~]$ timeout 5 bash -c '</dev/tcp/10.20.1.185/22' && echo "SSH PORT OPEN" || echo "SSH PORT UNREACHABLE"
SSH PORT UNREACHABLE
[ec2-user@ip-10-10-2-83 ~]$
~~~



10.2 Reverse Connectivity: App-B to App-A

Source: App-B (`10.20.1.185`)

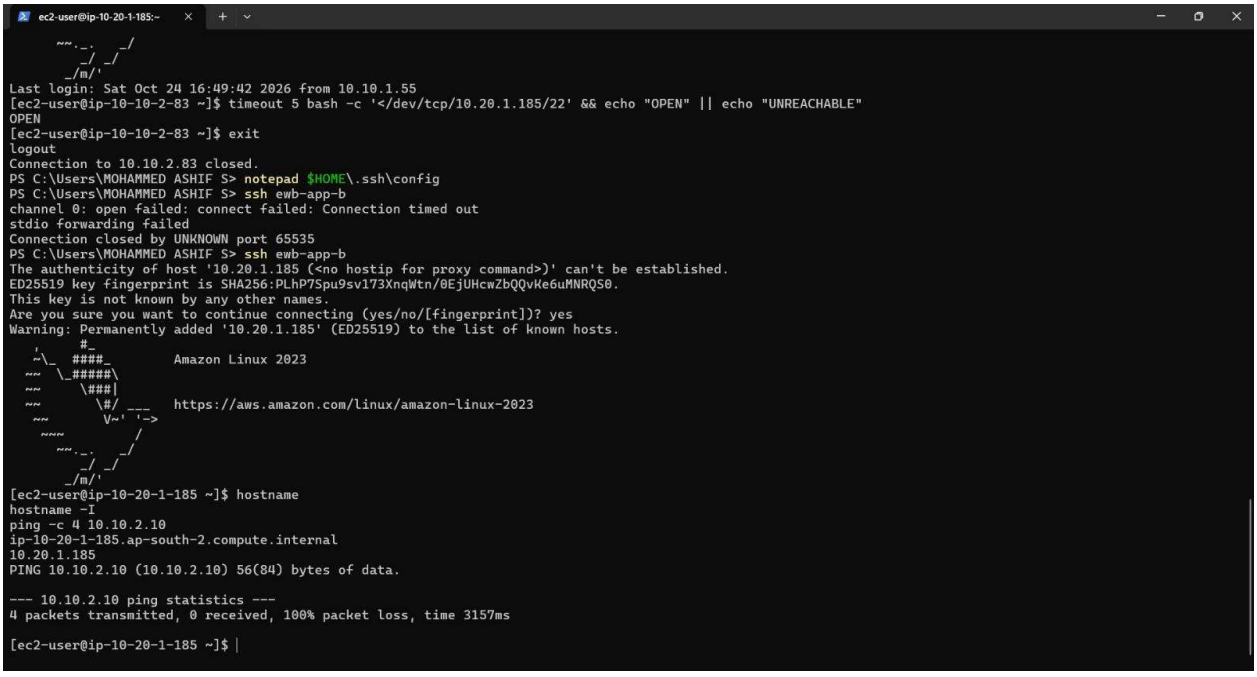
Destination: App-A (`10.10.2.83`)

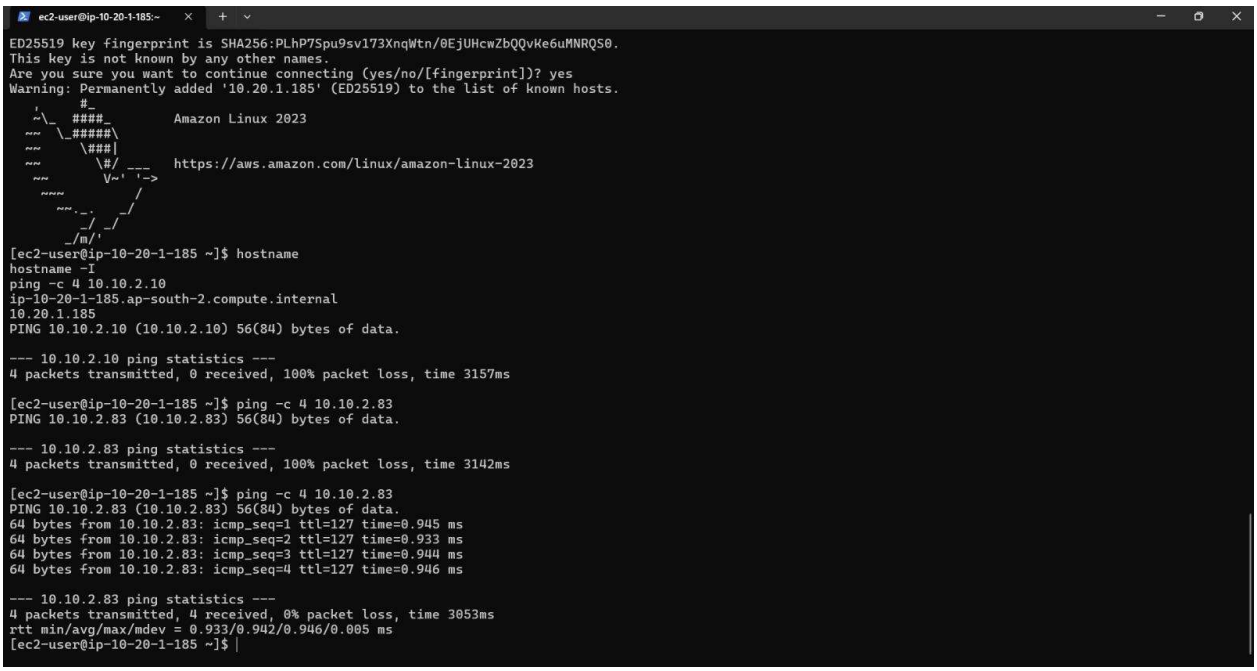
Command:

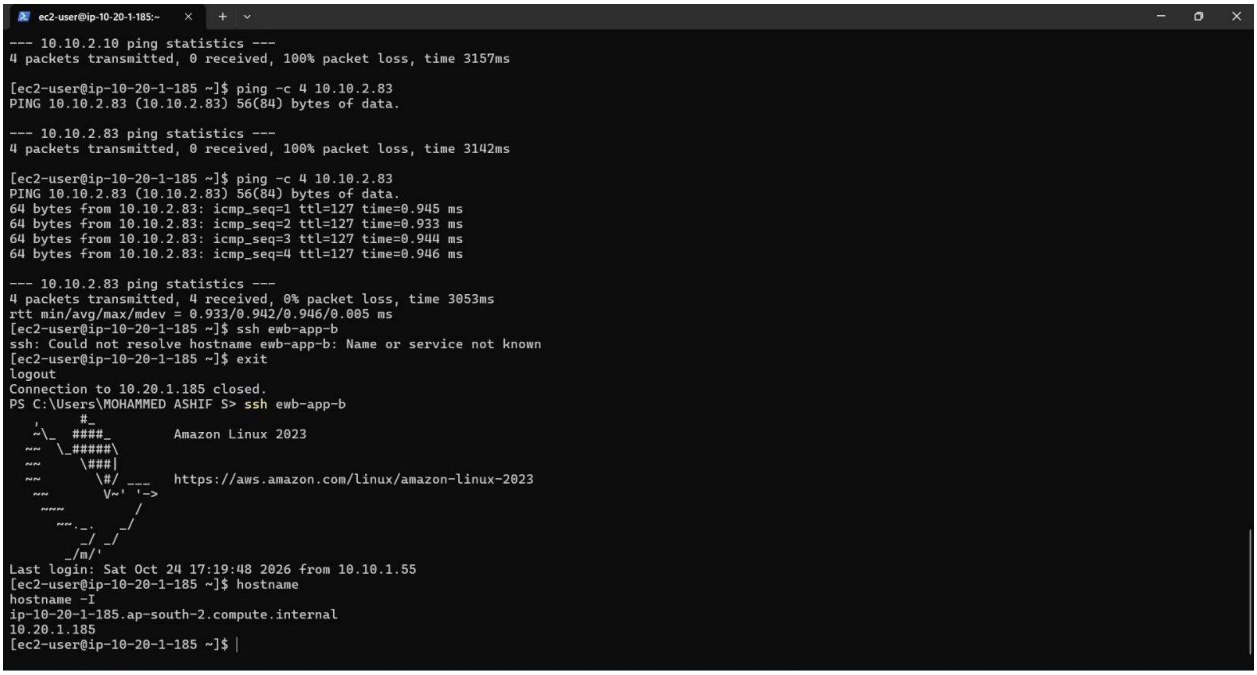
```
ping -c 4 10.10.2.83
```

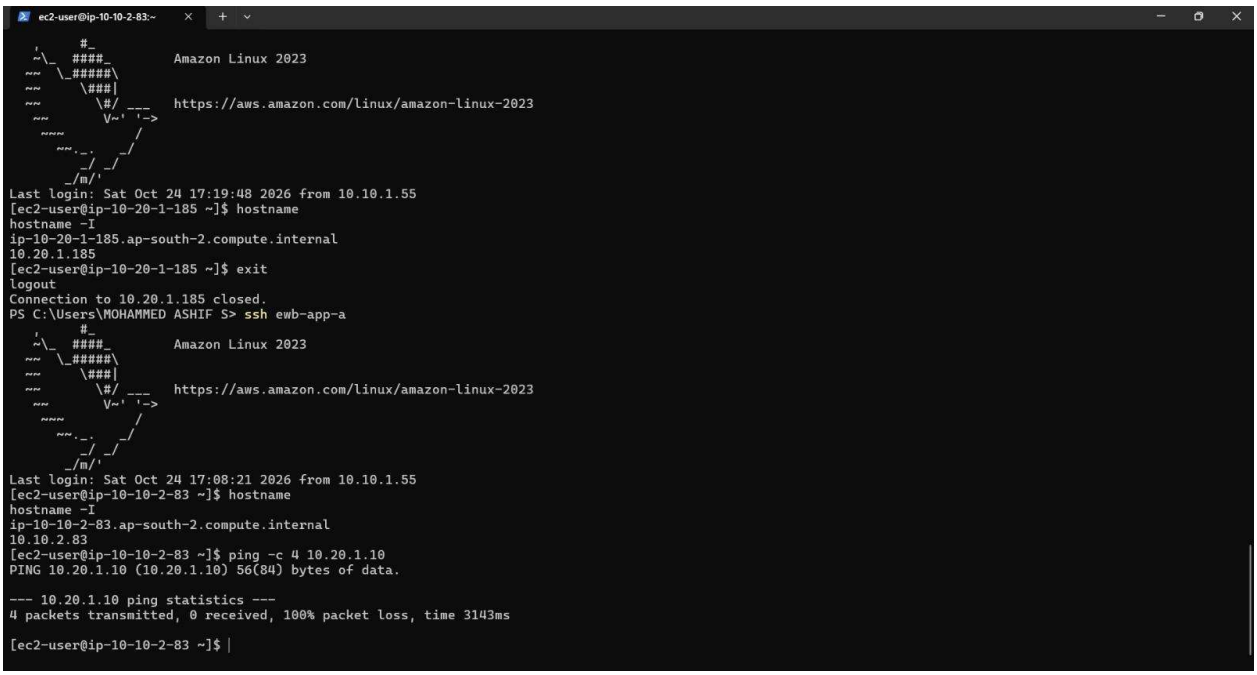
Observed result: Four packets were transmitted, four packets were received, and packet loss was 0%.

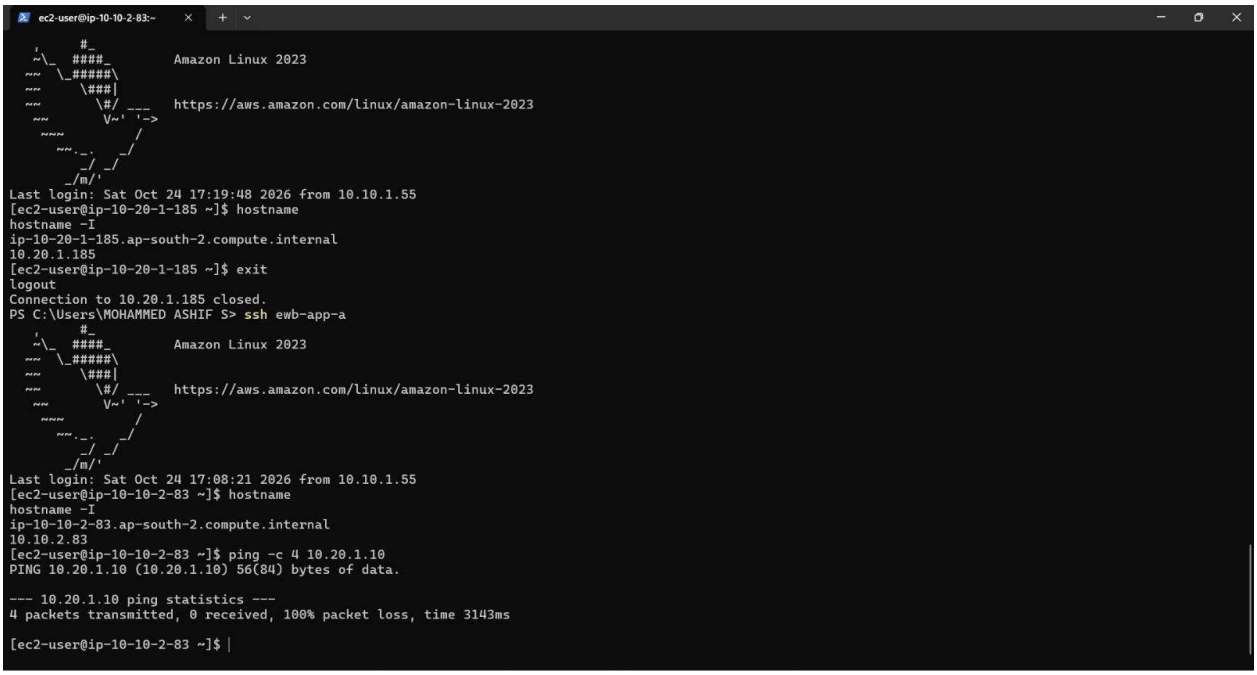
Conclusion: Reverse private connectivity was successful.

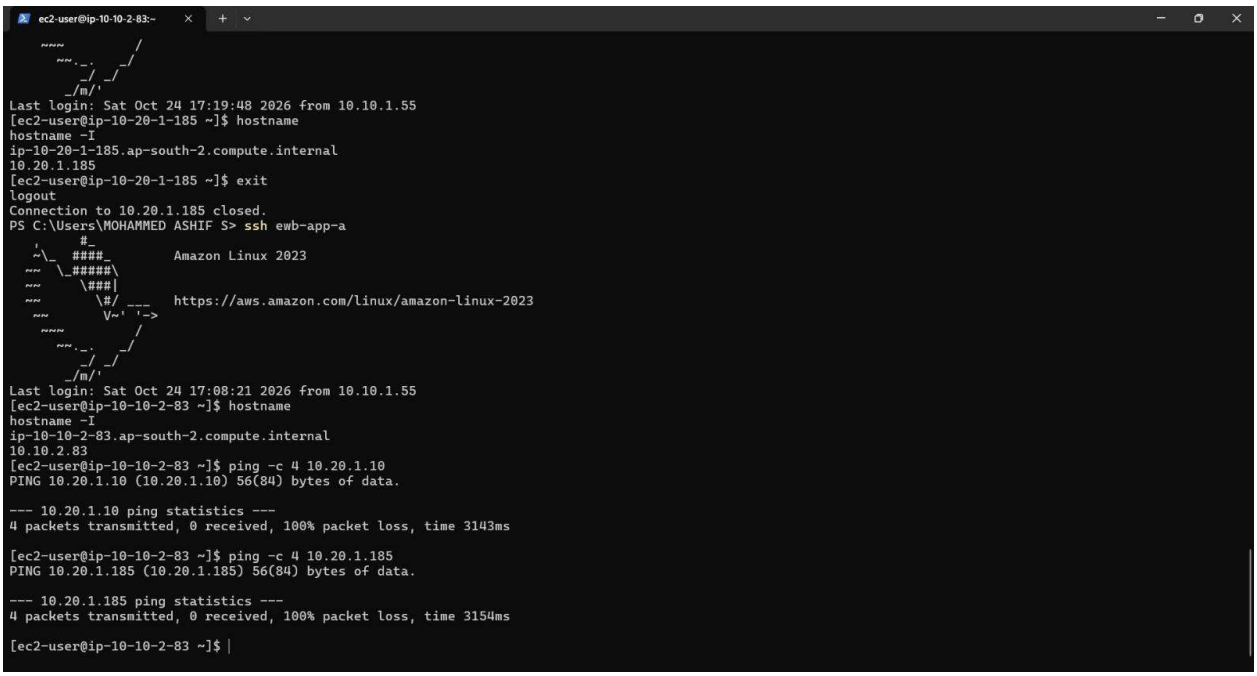

ec2-user@ip-10-20-1-185:~
Last login: Sat Oct 24 16:49:42 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ timeout 5 bash -c '</dev/tcp/10.20.1.185/22' && echo "OPEN" || echo "UNREACHABLE"
OPEN
[ec2-user@ip-10-10-2-83 ~]$ exit
logout
Connection to 10.10.2.83 closed.
PS C:\Users\MOHAMMED ASHIF S> notepad $HOME\.ssh\config
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-b
channel 0: open failed: connect failed: Connection timed out
stdio forwarding failed
Connection closed by UNKNOWN port 65535
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-b
The authenticity of host '10.20.1.185 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:PLhP7Spu9sv173XnqWtn/0EjUHcwZbQQvKe6uMNRQS0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.185' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-20-1-185 ~]$ hostname
hostname -I
ping -c 4 10.10.2.10
ip-10-20-1-185.ap-south-2.compute.internal
10.20.1.185
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3157ms
[ec2-user@ip-10-20-1-185 ~]$


ec2-user@ip-10-20-1-185:~
ED25519 key fingerprint is SHA256:PLhP7Spu9sv173XnqWtn/0EjUHcwZbQQvKe6uMNRQS0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.185' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-20-1-185 ~]$ hostname
hostname -I
ping -c 4 10.10.2.10
ip-10-20-1-185.ap-south-2.compute.internal
10.20.1.185
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3157ms
[ec2-user@ip-10-20-1-185 ~]$ ping -c 4 10.10.2.83
PING 10.10.2.83 (10.10.2.83) 56(84) bytes of data.
--- 10.10.2.83 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3142ms
[ec2-user@ip-10-20-1-185 ~]$ ping -c 4 10.10.2.83
PING 10.10.2.83 (10.10.2.83) 56(84) bytes of data.
64 bytes from 10.10.2.83: icmp_seq=1 ttl=127 time=0.945 ms
64 bytes from 10.10.2.83: icmp_seq=2 ttl=127 time=0.933 ms
64 bytes from 10.10.2.83: icmp_seq=3 ttl=127 time=0.944 ms
64 bytes from 10.10.2.83: icmp_seq=4 ttl=127 time=0.946 ms
--- 10.10.2.83 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3053ms
rtt min/avg/max/mdev = 0.933/0.942/0.946/0.005 ms
[ec2-user@ip-10-20-1-185 ~]$


ec2-user@ip-10-20-1-185:~
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3157ms
[ec2-user@ip-10-20-1-185 ~]$ ping -c 4 10.10.2.83
PING 10.10.2.83 (10.10.2.83) 56(84) bytes of data.
--- 10.10.2.83 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3142ms
[ec2-user@ip-10-20-1-185 ~]$ ping -c 4 10.10.2.83
PING 10.10.2.83 (10.10.2.83) 56(84) bytes of data.
64 bytes from 10.10.2.83: icmp_seq=1 ttl=127 time=0.945 ms
64 bytes from 10.10.2.83: icmp_seq=2 ttl=127 time=0.933 ms
64 bytes from 10.10.2.83: icmp_seq=3 ttl=127 time=0.944 ms
64 bytes from 10.10.2.83: icmp_seq=4 ttl=127 time=0.946 ms
--- 10.10.2.83 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3053ms
rtt min/avg/max/mdev = 0.933/0.942/0.946/0.005 ms
[ec2-user@ip-10-20-1-185 ~]$ ssh ewb-app-b
ssh: Could not resolve hostname ewb-app-b: Name or service not known
[ec2-user@ip-10-20-1-185 ~]$ exit
logout
Connection to 10.20.1.185 closed.
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-b
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sat Oct 24 17:19:48 2026 from 10.10.1.55
[ec2-user@ip-10-20-1-185 ~]$ hostname
hostname -I
ip-10-20-1-185.ap-south-2.compute.internal
10.20.1.185
[ec2-user@ip-10-20-1-185 ~]$


ec2-user@ip-10-10-2-83:~
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sat Oct 24 17:19:48 2026 from 10.10.1.55
[ec2-user@ip-10-20-1-185 ~]$ hostname
hostname -I
ip-10-20-1-185.ap-south-2.compute.internal
10.20.1.185
[ec2-user@ip-10-20-1-185 ~]$ exit
logout
Connection to 10.20.1.185 closed.
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-a
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sat Oct 24 17:08:21 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ hostname
hostname -I
ip-10-10-2-83.ap-south-2.compute.internal
10.10.2.83
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3143ms
[ec2-user@ip-10-10-2-83 ~]$

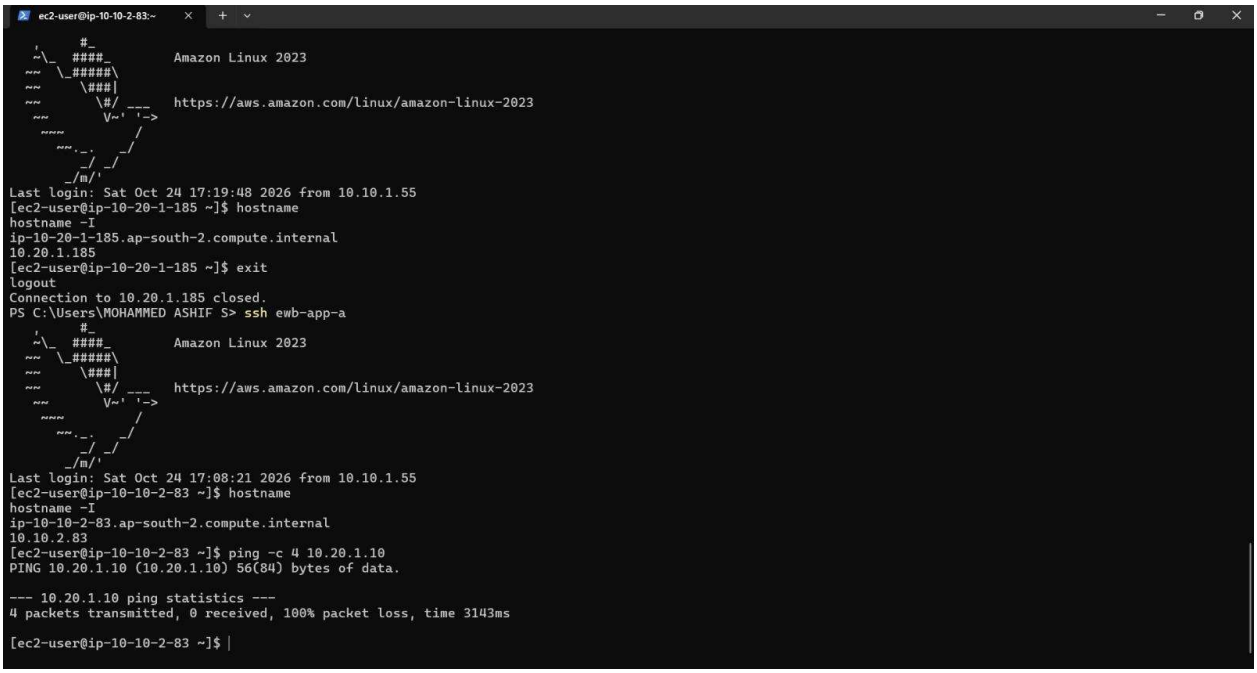


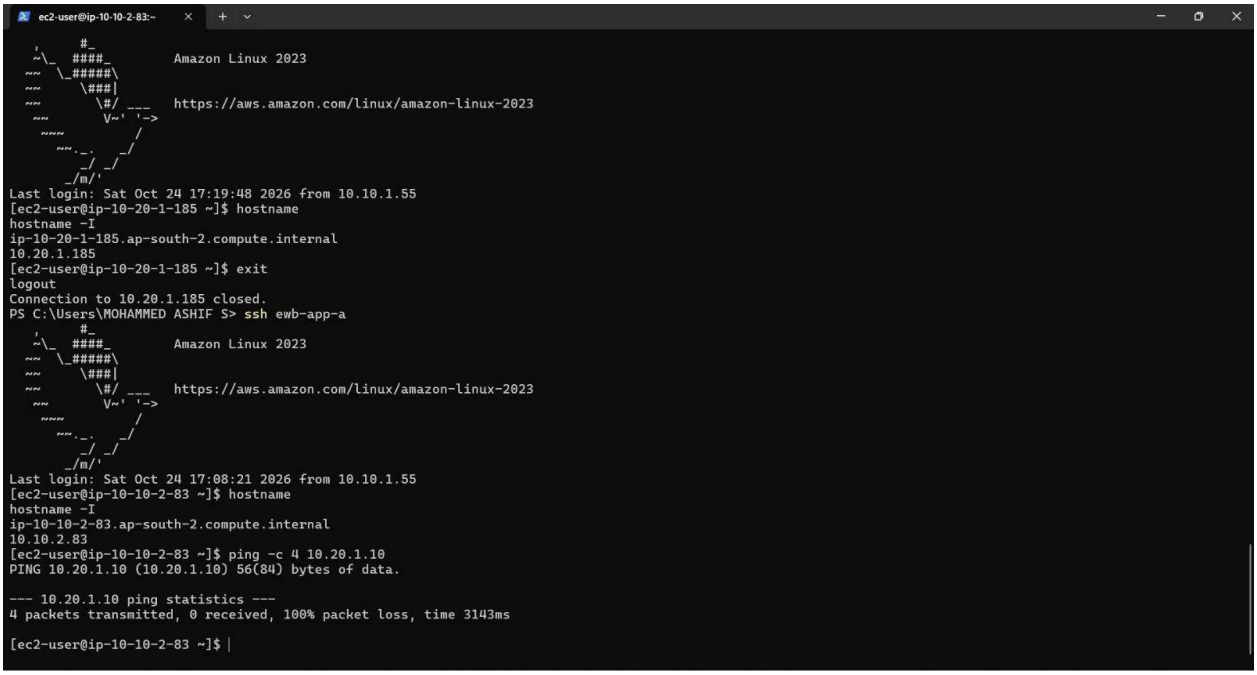


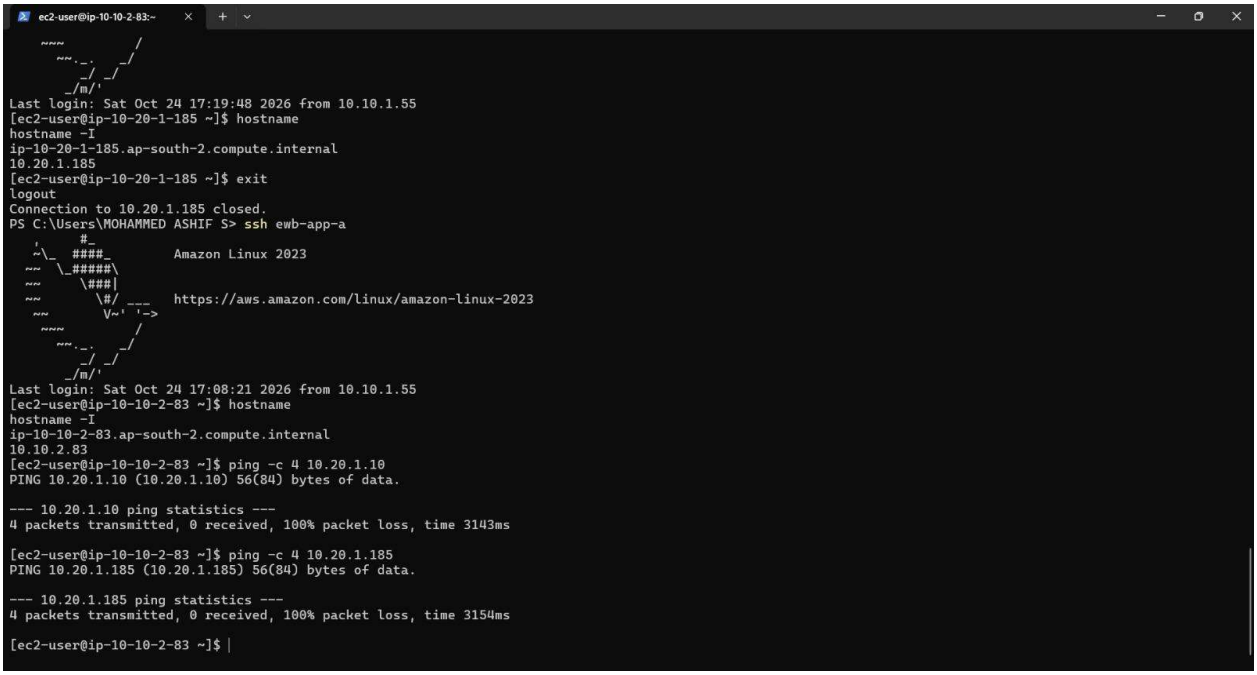
10.3 SSH Access to App-B

The command `ssh ewb-app-b` was executed from the local computer using the configured bastion host as the SSH jump host.

The terminal successfully opened a shell on App-B, whose hostname identified the private instance `ip-10-20-1-185`.


ec2-user@ip-10-10-2-83:~
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sat Oct 24 17:19:48 2026 from 10.10.1.55
[ec2-user@ip-10-20-1-185 ~]$ hostname
hostname -I
ip-10-20-1-185.ap-south-2.compute.internal
10.20.1.185
[ec2-user@ip-10-20-1-185 ~]$ exit
logout
Connection to 10.20.1.185 closed.
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-a
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sat Oct 24 17:08:21 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ hostname
hostname -I
ip-10-10-2-83.ap-south-2.compute.internal
10.10.2.83
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3143ms
[ec2-user@ip-10-10-2-83 ~]$


ec2-user@ip-10-10-2-83:~
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sat Oct 24 17:19:48 2026 from 10.10.1.55
[ec2-user@ip-10-20-1-185 ~]$ hostname
hostname -I
ip-10-20-1-185.ap-south-2.compute.internal
10.20.1.185
[ec2-user@ip-10-20-1-185 ~]$ exit
logout
Connection to 10.20.1.185 closed.
PS C:\Users\MOHAMMED ASHIF S> ssh ewb-app-a
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Sat Oct 24 17:08:21 2026 from 10.10.1.55
[ec2-user@ip-10-10-2-83 ~]$ hostname
hostname -I
ip-10-10-2-83.ap-south-2.compute.internal
10.10.2.83
[ec2-user@ip-10-10-2-83 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3143ms
[ec2-user@ip-10-10-2-83 ~]$

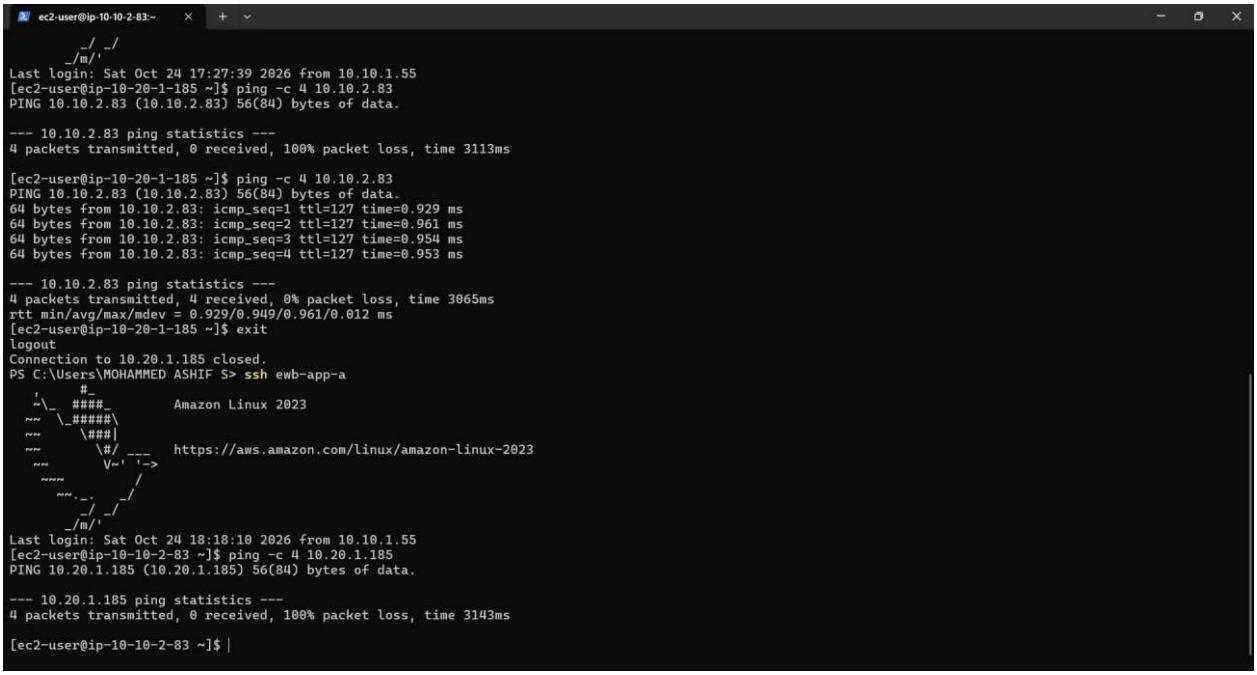


10.4 Controlled Failure Test

The inbound ICMP rule permitting traffic from App-A was temporarily removed from App-B's security group.

The forward ping test then returned four transmitted packets with 100% packet loss.

Conclusion: The test demonstrated that the security group rule was required for the tested ICMP connectivity.

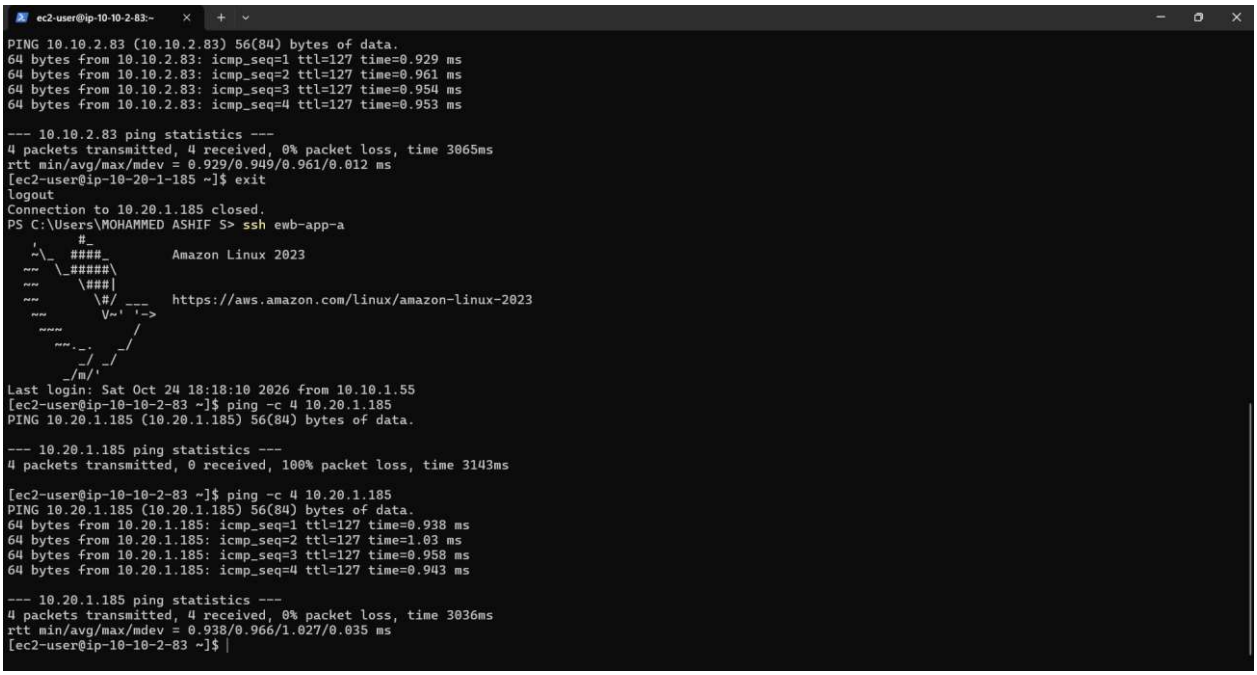


10.5 Recovery Test

The original inbound ICMP rule was restored with App-A's actual private IP address, `10.10.2.83/32`.

The forward ping test was repeated and returned four received packets with 0% packet loss.

Conclusion: Connectivity was restored after the security group rule was restored.



11. Baseline Test

The workbook requires a failed connectivity test from App-A before the peering connection and peer routes are configured.

Expected result: No ping replies before peering and peer routes are configured.

Evidence status: Baseline Connectivity Test

The baseline test was intended to verify that App-A could not communicate with App-B before VPC peering was configured. However, a valid screenshot confirming the initial test could not be verified in the available evidence. After configuring VPC peering, successful connectivity between the two instances was confirmed.

12. Learning Outcomes

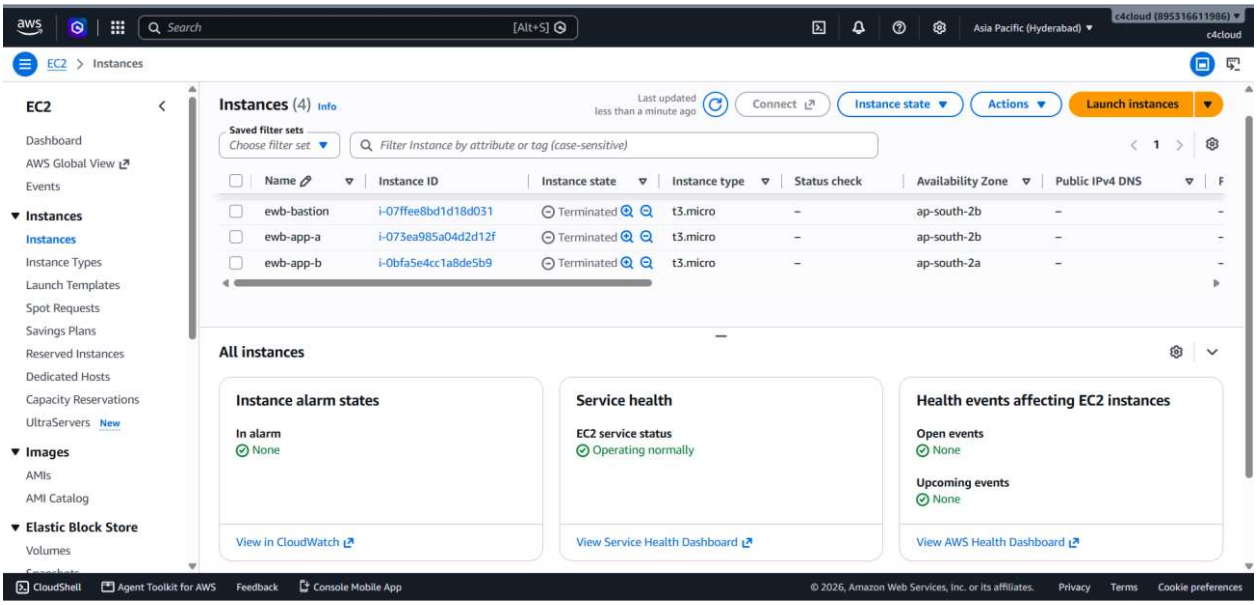
Through this project, I gained practical experience in creating VPCs and subnets, configuring route tables, establishing VPC Peering, and controlling network traffic with security groups.

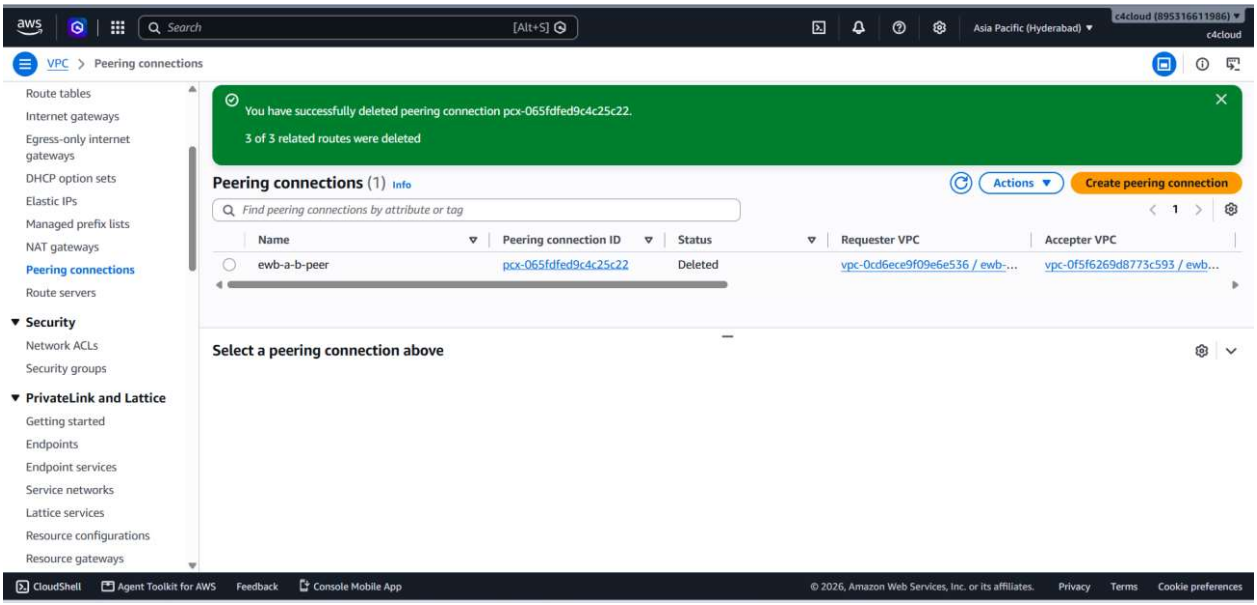
I learned that a peering connection alone does not guarantee connectivity. Correct routes are required on both sides, and security groups must allow the necessary traffic. I also learned how to verify connectivity using ping, connect to a private EC2 instance through a bastion host, and troubleshoot network access through a controlled failure and recovery test.

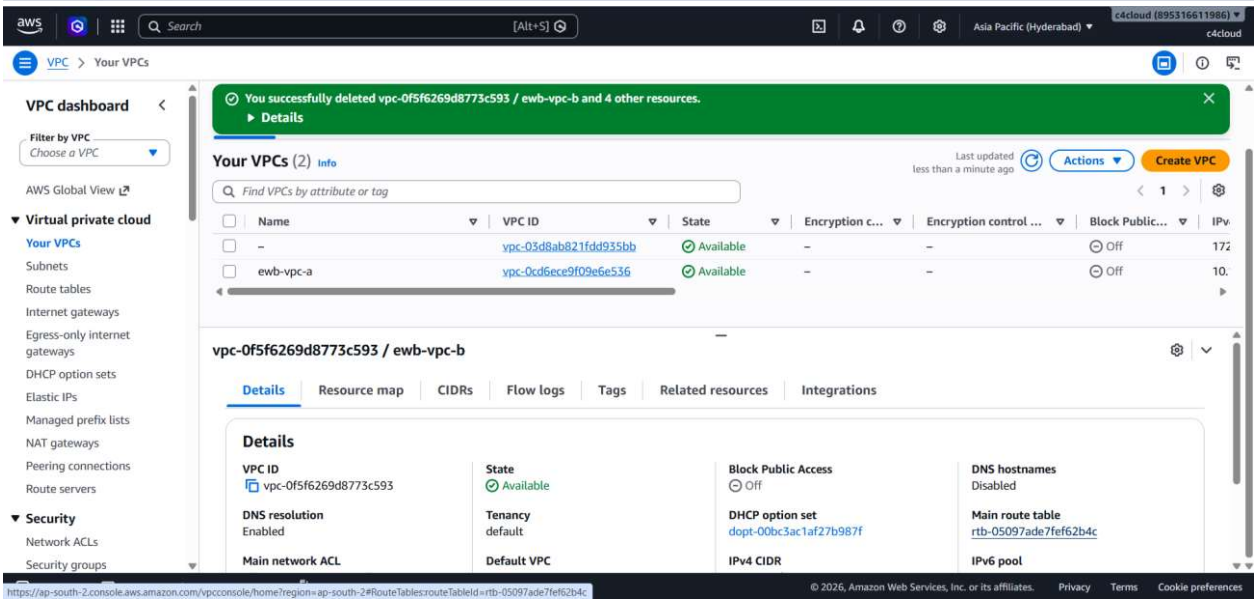
13. Cleanup and Cost Management

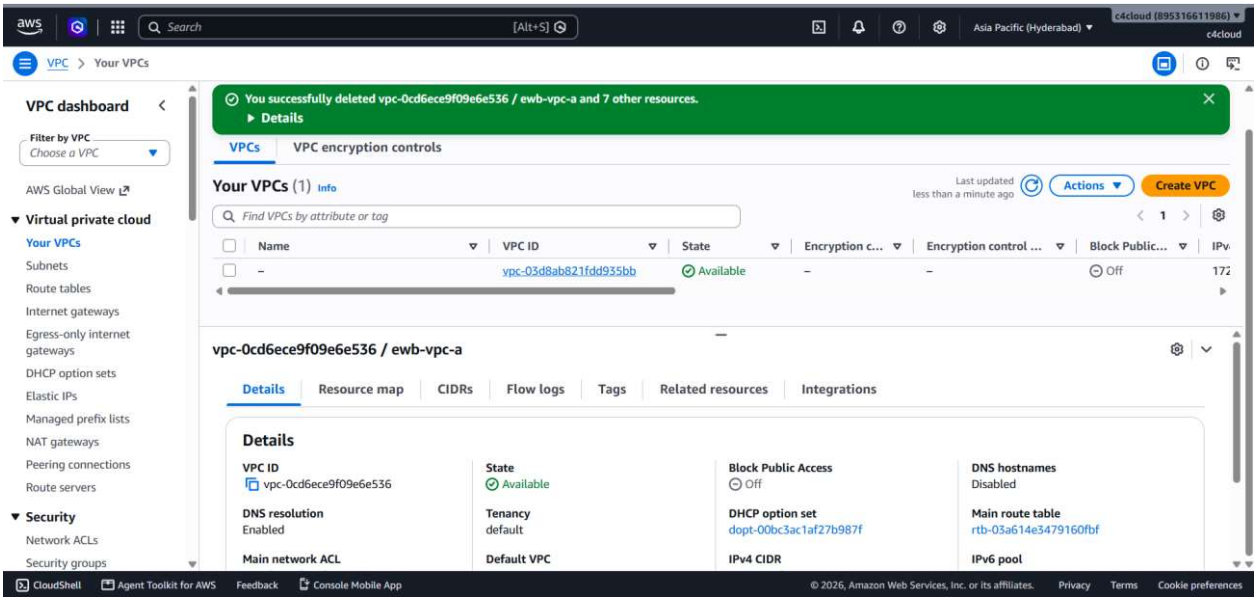
After saving the required screenshots and completing the report, the lab resources should be removed according to the workbook cleanup procedure.

Cleanup evidence should be added after the EC2 instances and remaining lab resources have actually been removed and verified.




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Egress-only internet gateways
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You have successfully deleted peering connection pcx-065fdfed9c4c25c22.
3 of 3 related routes were deleted
Peering connections (1) Info
Actions
Create peering connection
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Name
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Status
Requester VPC
Accepter VPC
ewb-a-b-peer
pcx-065fdfed9c4c25c22
Deleted
vpc-0cd6ece9f09e6e536 / ewb-...
vpc-0f5f6269d8773c593 / ewb...
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Agent Toolkit for AWS
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Peering connections
Route servers
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Security groups
You successfully deleted vpc-0f5f6269d8773c593 / ewb-vpc-b and 4 other resources.
Details
Your VPCs (2) Info
Last updated less than a minute ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv
–
vpc-03d8ab821fdd935bb
Available
Off
ewb-vpc-a
vpc-0cd6ece9f09e6e536
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Off
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Disabled
DNS resolution
Enabled
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default
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dopt-00bc3ac1af27b987f
Main route table
rtb-05097ade7fef62b4c
Main network ACL
Default VPC
IPv4 CIDR
IPv6 pool
https://ap-south-2.console.aws.amazon.com/vpcconsole/home?region=ap-south-2#RouteTables:routeTableId=rtb-05097ade7fef62b4c
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14. Conclusion

The project demonstrated private communication between two AWS VPCs using VPC Peering. The route tables provided the forward and return paths, and security groups controlled the permitted traffic.

The successful forward and reverse ping tests confirmed private connectivity between App-A and App-B. The SSH test demonstrated administrative access through the bastion host. The controlled failure and recovery tests demonstrated the effect of security group rules on network connectivity.

This hands-on project improved my understanding of AWS networking, private IP communication, route tables, security groups, SSH access, and troubleshooting.

End of Report